

Internal Migration and Labor Market Adjustments in the Presence of Non-wage Compensation

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Abstract

In this paper, we argue that adjustments in non-wage compensation are empirically relevant and thus can have important implications for studying the effects of labor supply shocks. We investigate the labor market impacts of internal migration in Brazil using a shift-share approach that combines weather-induced migration with past settlement patterns in each destination. Increasing migration inflows reduces formal employment and increases informal jobs by a similar magnitude. Consistent with downward wage rigidity, we find a weaker negative effect on formal earnings than on informal and a negative impact on the share of workers receiving non-wage benefits. As a result, total compensation falls. Less educated individuals bear most of these costs. Unemployment and labor force participation increase, mainly driven by non-head members of the household. We interpret our findings within a simple model with two sectors in an economy with different levels of intersectoral linkages and with flexible or fixed benefits that generates predictions broadly consistent with our findings.

Keywords: Internal migration, wages, employment, non-wage benefits.

JEL Codes: J2, J3, J61, O15.

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1 Introduction

Migration within and beyond borders has important implications regarding development, demographic and economic dynamics. As a response, an extensive literature on the impacts of migration on the native population has emerged, particularly in terms of employment and wage levels. In a recent book, [Borjas \(2014\)](#) summarizes his vast contribution to the field and underscores the costs of immigration for competing native workers. On the other hand, a growing fraction of scholars has concluded that migration may have more nuanced effects ([Card and Peri, 2016](#)). [Card \(2009\)](#) finds that immigration to the United States has only a minor impact on native wages, and [Ottaviano and Peri \(2012\)](#) report minor positive wage effects.

Canonical partial equilibrium models with perfect competition and substitution between natives and migrants predict full adjustment through wages when natives are immobile or lower native employment when wages are rigid (for an early example, see [Altonji and Card, 1991](#)). Attempts to reconcile the apparently contradicting empirical evidence include expanding models to accommodate multiple outputs and technology margins ([Lewis, 2011](#); [Dustmann and Glitz, 2015](#)) as well as recognizing that different specifications measure different parameters ([Dustmann et al., 2016](#)).

While the debate remains contentious, implicit in this discussion is a common but under-considered assumption that non-wage aspects of jobs are fixed. Allowing for adjustments along these margins may have important implications for the study of labor markets. [Clemens \(2021\)](#) argues they may explain existing controversies over the economics of minimum wages. Less is known about the role of non-wage adjustments in our understanding of the consequences of migration.

In this paper, we argue that adjustments in non-wage compensation are empirically relevant and thus can have important implications for studying the effects of labor supply shocks due to migration. In particular, we investigate the impacts of internal migration in Brazil on the labor market outcomes of natives in a setting where downward wage rigidity is present, non-wage benefits are a significant margin of compensation, and labor informality is pervasive. This setup allows us to study how firms and workers – when adjusting to a labor supply shock due to increased migration inflows – may circumvent the binding minimum wages by reducing non-wage benefits of formal jobs or simply lowering salaries in unregulated informal markets.

The theory is based on a simple model that generates predictions for the impact of migration on labor markets with two sectors in an economy with different levels of intersectoral linkages and with endogenous or fixed benefits. From low to medium levels of linkages, the impact of migration in terms of employment and wage drops in magnitude. Allowing firms to adjust benefits as a response to shocks also softens the impact on employment. Non-wage benefits are a relevant margin of adjustment for firms, especially in more regulated labor markets. They ease constraints and allow

employers to partially absorb shocks.

Brazil provides a good environment for our investigation for three reasons. First, over 3 million people in the Brazilian Semi-arid, a historical source of climate migrants, left their hometowns during our sample period of 1996-2010. Second, a within-country analysis minimizes econometric concerns about allocating migrants to particular skill groups (Dustmann et al., 2012). Third, over 40 percent of workers are employed in the less frictional informal labor sector, where firms do not comply with labor market statutes, such as minimum wage laws and layoff regulations. The rest of the workforce participates in the formal sector where the minimum wage is binding (above 70% of the median wage) and non-wage compensation is frequently offered. Indeed, over 31 million people, or 20% of registered workers, are covered through employer-provided health insurance. After payroll expenses, this is the second highest component of total labor costs (ANS, 2019). Also, 40% of these workers receive food subsidies, costing firms about 57% of the minimum wage per worker.¹ To the extent that workers value non-wage benefits, changes in this margin of adjustment can have important welfare implications.

To address the econometric concerns associated with the fact that migrants tend to move to areas with better labor market opportunities, we take advantage of a recent body of work that provides a clear framework for distinguishing sufficient conditions for identification and properly computing standard errors (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2021; Jaeger et al., 2018; Adao et al., 2019). In particular, we combine two extensively used identification strategies into a shift-share instrument approach. First, we exploit exogenous rainfall and temperature shocks (or “shift”) at the origin to predict the number of individuals leaving each Semi-arid’s municipality. Then we leverage the history of the Semi-arid as a large source of climate migrants and use the past settlement patterns (or “share”) to allocate migration outflows to destination areas (Munshi, 2003; Boustan et al., 2010). The resulting predicted inflow of migrants is an instrument for observed migration.

Our results show that increasing the rate of migration inflows by one percentage point reduces the share of formal employment among native workers by 0.13*p.p.* while increasing the number of informal² jobs by 0.11*p.p.*. These results are consistent with a binding minimum wage such that migration shocks lead to lower formal employment as formal sector employers cannot adjust wages downward, and individuals who lost their formal jobs being absorbed by informal firms or self-employment, which are more competitive labor sectors. Thus the overall effect on native total employment across sectors is small or even null.

Regarding compensation, we find a decrease between 0.59% and 1.00% on average earnings in the formal sector and a negative impact on the share of formal workers receiving employer-provided health insurance in the range of 0.31*p.p.* to 0.47*p.p.*, food

¹Arbache and Ferreira (2001) based on various sources estimate the average cost of providing some job benefits in Brazil.

²Our definition of informal sector also includes self-employed workers.

vouchers from 0.33*p.p.* to 0.69*p.p.* and transportation subsidies from 0.37*p.p.* to 0.57*p.p.* Our evidence on employer-sponsored health insurance provision is complemented with firm-level administrative data on health insurance contracts matched to firm-level data on formal sector jobs. We find that firms operating in a municipality that receives more incoming migrants are less likely to provide health insurance to employees, an effect that is mostly driven by large firms. Despite declines in the provision of non-monetary benefits, which increases labor demand, employment in the formal sector still drops.³ Wages in the formal sector reduce across the entire wage distribution but more so for higher wage percentiles which is consistent with binding minimum wages.

For individuals employed in the informal sector or self-employed we show a decrease on earnings between 0.75% and 0.99% mostly concentrated on the bottom third of the wage distribution, consistent with predictions from a two-sector labor market model where wages can freely adjust in the informal sector, and given a less educated migrant workforce that increases competition relatively more among informal workers.

We provide a simple back-of-the-envelope exercise to gauge whether changes in non-wage compensation are important relative to the changes in earnings. When accounting for food subsidies and private health insurance, a 1*p.p.* increase in the number of migrants reduces *total compensation* by 1.06%. This highlights that, if we do not account for the non-wage compensation margin of adjustment, we will underestimate the change in overall compensation by 80% percent, assuming that workers value such benefits at their cost estimate.

To explore the mechanics behind our results, we investigate whether municipalities where the national minimum wage was more binding at baseline had stronger employment and/or nonwage compensation adjustments. We find that in places with greater minimum wage bite, the adjustment occurs mostly via informal earnings and formal employment. In regions where the minimum wage is less binding, we find little effect on informal employment and earnings. The adjustment in the formal sector occurs in total compensation - both on earnings and non-wage benefits.

Heterogeneity analysis shows that the estimated effects are stronger for less educated native workers, which is consistent with the fact that they directly compete with Semi-arid's migrants. When compared to those with high education, less educated natives are more likely to exit the formal sector and experience a 26% greater wage reduction. Moreover, as more low-education workers earn close or equal to the minimum wage, the negative impact on the most frequently non-wage benefits provided by firms is greater for them. This suggests that welfare declines more for low-income workers, therefore, rising welfare inequality among natives.

Next, we find that unemployment and labor force participation increases by roughly 0.07 – 0.09*p.p.*, which may seem at odds with previous results since earnings fall in the informal sector and benefits drop in the formal sector. By running separate regressions

³Recent literature, as discussed in [Clemens \(2021\)](#), reports a negative correlation between minimum wage increases and health insurance provision, with the variation of this benefit offsetting about 15% of the cost with minimum wage increases.

for head and non-head of the household, we find that almost all the impact on the employment margins comes from the head of household while the change in unemployment and inactivity rates are led by the non-head member, consistent with the added worker effect (Lundberg, 1985).⁴

We then turn our attention to the long-term impacts of migration on local labor markets in Brazil. Our results indicate that the estimated effects on average earnings in the formal sector remain mostly the same, but in the informal sector decrease even further. In the case of employment, we see a larger negative impact on formal workers while there are no significant effects on informal jobs. As for non-wage benefits, the impact on health insurance is mostly the same as in the short run, while the negative effects on transport subsidies are larger and there is no significant effect on food benefits. Also, we show that a potential mechanism behind these dynamics is that short-run effects might be partially offset by further internal migration as (mainly low-education) natives respond to the adverse effects by moving to markets that were not directly targeted by migrant arrivals.

Our work is related to a broad literature that examines the impact of migration flows on labor market outcomes of natives (see Borjas, 2014 and Dustmann et al., 2016 for a review). Despite the fact that migration within countries is a larger phenomenon,⁵ most studies are concerned with international immigration to high-income countries, with particular attention given to Mexican immigration to the United States (Borjas, 2003) and, more recently, to immigration to Western Europe (Dustmann et al., 2012). Some of these studies find that the wages of natives are harmed by immigration (Borjas and Monras, 2017), while others find only a minor negative effect on native wages (Card, 2001), or even positive (Ottaviano et al., 2013; Foged and Peri, 2016; Azoulay et al., 2022).⁶ A smaller set of studies explore environmental and other economic shocks to study the causal impact of internal migration on local labor markets in the US (Boustan et al., 2010; Hornbeck, 2012; Imbert et al., 2022).⁷ More closely related to our work is Kleemans and Magruder (2018) who study the impacts of internal migration in a developing country, Indonesia, from a two-sector labor market perspective. They show that internal migration reduces employment in the formal sector and earnings in the informal sector.⁸

⁴The “added worker effect” in a broader sense here refers to an increase in the labor supply of secondary earners (typically wives and children) when the primary earner (husbands) becomes unemployed or lose a formal sector job where benefits, sometimes extended to the family, are provided.

⁵Rough estimates indicate that global internal migration sits around 740 million (UNDP, 2009), approximately three times the estimated number of international migrants (UN DESA, 2017).

⁶Dustmann et al. (2016) argue that such often contradictory estimates are a result of (i) different empirical specifications (sources of variation), as well as the fact that labor supply elasticity differs across different groups of natives, and immigrants and native do not compete in the labor market within the same education-experience cells.

⁷See also Molloy et al. (2011) for a comprehensive literature review on the determinants of internal migration in the U.S. and Lagakos (2020) on urban-rural internal movements.

⁸This approach relates to the seminal work of Harris and Todaro (1970). A similar extension and test of this model is provided in Busso et al. (2021) using census data from Brazil.

Our contribution to the economics of migration literature is fourfold. First, we show that firms systematically adjust non-wage benefits in response to labor supply shocks. Second, accounting for such adjustments are key to understanding the effects of migration on natives. Third, we provide evidence on the effects of internal migration on local labor markets in a large developing country and show that these different adjustment patterns are relevant even in the presence of informality. Fourth, we add to a growing body of evidence that migration is a relevant coping mechanism against climate change, especially for vulnerable populations in rural areas of developing countries (Skoufias et al., 2013; Assunção and Chein, 2016).

Non-wage benefits are also an important part of compensation in developed countries. In the US, employer-provided health insurance and other benefits account for around one-third of compensation costs (Clemens et al., 2018). 74% of firms in Europe paid non-base wage components such as benefits and bonuses in 2013 (Babecký et al., 2019). Evidence shows that firms adjust non-wage components when facing adverse economic shocks (Babecký et al., 2019) or as a strategy to offset collective bargaining (Cardoso and Portugal, 2005), particularly when base wages are rigid (Babecký et al., 2012). We add to this literature by showing that non-wage benefits are an important margin of adjustment in the case of labor supply shocks due to internal migration.

This paper is organized as follows. In the next section, we first present background information on the Brazilian Semiarid region and local labor markets. Section 3 outlines a simple framework for interpreting our findings. Section 4 describes the data and empirical framework, and reports first-stage estimates that link observed migration patterns to our predicted migration flows. Next, we present and analyze the main results on employment, wages and non-wage wage benefits in Section 5. We also study the sensitivity and heterogeneity of our main estimates. Finally, we interpret our main estimates in light of our simple model and conclude.

2 Background

In this section, we first describe the economic background and weather conditions in the Semiarid region, the functioning of local labor markets in Brazil, and a simple framework in an effort to contextualize our analysis. We then discuss the main sources of data regarding labor market outcomes, migration flows, and weather, and present some descriptive statistics.

2.1 Brazilian Semiarid

The Brazilian Semiarid encompasses 960 municipalities spread over 9 states, covering an area of around 976,000km².⁹ According to the official definition by the Ministry of

⁹That is roughly the same as the territory of Germany and France combined. The semiarid comprises 11 percent of the Brazilian territory and includes parts of almost all Northeastern states (except for *Maranhão*) plus the northern area of *Minas Gerais*, but it does not cover any state capital.

National Integration, a municipality qualifies as Semiarid if at least one of these three criteria holds: (i) annual average precipitation below 800 mm between 1961 and 1990; (ii) aridity index up to 0.5¹⁰; (iii) risk of drought above 60%¹¹. The average historical precipitation in the Semiarid is about 780mm, as opposed to around 1,600 mm for the rest of the country¹², while average temperature is around 25°C. The rainy season occurs between November and April, with the highest levels of precipitation after February, when the sowing seasons typically starts.

Municipalities are relatively small with median population around 20,000 and have economies mainly based on agriculture and cattle ranching in small subsistence properties. Local economic activity is particularly susceptible to weather shocks (Wang et al., 2004), with some studies showing a loss of up to 80% of agricultural production in periods of long drought (Kahn and Campus, 1992). About 80% of the children lived below the poverty line and infant mortality reached 31 per 1,000 births in 1996, compared to a national average of 25% and 15 per 1,000 births, respectively (Rocha and Soares, 2015). More than 80% of the adult population had less than 8 years of schooling in 1991.

Such poor socioeconomic indicators associated with periods of extreme drought have historically driven large outflows of migrants - or so-called *retirantes* - from the Semiarid to other areas of the country (Barbieri et al., 2010). During the 1960s and 1970s, net migration out of Northeastern states (where most of the Semiarid is located) was 2.2 and 3.0 million individuals (Carvalho and Garcia, 2002), which correspond to net migration rates of -7.6 and -8.7%, respectively. Between 1996 and 2010, around 3.0 million people left the Semiarid alone searching for better conditions elsewhere in the country. Appendix Figure B1 shows that these migrants tend to be historically concentrated in some states. São Paulo alone harbored over 30 percent of the people arriving from the Semiarid in the last four decades. However, in relative terms, incoming migrants represented a population increase of above 2% for the top 10 receiving states.

2.2 Labor Markets in Brazil

A common feature of labor markets in developing countries is the existence of a two-sector economy where the informal sector accounts for one to two-thirds of the GDP (see Perry et al. (2007) and Ulyssea (2020) for a review). In Brazil, over 40% of individuals work in the informal sector (those without registration or who do not contribute to social security) including the majority of the self-employed who are not protected through social security. When firms hire workers under a formal contract they are subject to several legal obligations, such as paying minimum wages and complying with safety regulations. Registration also entitles workers to other benefits such as a wage contract, which in Brazil prevents downward adjustment, working up to 44 hours weekly, paid annual leave, paternity or maternity leave, retirement pension,

¹⁰Thornthwaite Index, which combines humidity and aridity for a given area, in the same period.

¹¹Defined as the share of days under hydric deficit, using the period 1970-1990.

¹²See Figure 11.

unemployment insurance, and severance payments (e.g. [Gonzaga, 2003](#); [Almeida and Carneiro, 2012](#); [Meghir et al., 2015](#); [Narita, 2020](#)).

If firms do not comply with working regulations they may be caught by the labor authorities and have to pay a fine. For example, a firm is fined about one minimum wage for each worker that is found unregistered, or the firm can be fined up to a third of a minimum wage per employee if it does not comply with mandatory contributions to the severance fund ([Almeida and Carneiro, 2012](#)).¹³ On the other hand, it is a well-known fact that compliant (formal) firms are those more visible to labor inspectors and thus subject to more inspections whereas informal firms are smaller and thus difficult to get caught ([Cardoso and Lage, 2006](#)). There are also other expected costs for formal firms associated with labor courts in case the worker is fired and decides to file a lawsuit against the firm. Judges decide in favor of workers in nearly 80% of cases ([Corbi et al., 2022](#)). This all points to a significant cost of operating in the formal sector, particularly for smaller firms. Imperfect enforcement and costly regulation are associated with high labor informality in the country.

Finally, as there is a strong overlap between the productivity distributions of formal and informal sectors ([Meghir et al., 2015](#)), even for lower percentiles of the overall distribution, both sectors should be affected by the influx of migrants. In other words, both sectors have workers who are close substitutes to the migrant workforce and thus will experience competition.

Non-wage compensation. In our empirical analysis, we focus on three main fringe benefits we observe in the data: private health insurance, food, and transport subsidies. In Brazil, benefits became popular in the 1980s, as the provision of food subsidies and employer-provided health insurance became more frequent among private sector firms ([Arbache, 1995](#)). Data from PNAD surveys for 1996-2009 indicate that 39% of workers in the formal sector receive food subsidies, 36% receive transport subsidies and 21% get private health insurance through their employers. [Arbache and Ferreira \(2001\)](#) estimate that benefits like food subsidies for instance cost around 57% of one minimum wage (around 16% of average total compensation). Similarly, Brazilian Federal Health Agency data ([ANS, 2018](#)) show that employer-provided health insurance cost on average R\$582 in 2018, which is 17% of total compensation in that same year. These numbers imply that depending on how firms opt to mix benefits in the workers' package, these expenses may add up to above 30% of the total payroll cost. In the US, benefits including employer-provided health insurance account for around one-third of compensation costs ([Clemens, 2021](#)).

There are at least two reasons that can explain the use of fringe benefits in workers' compensation. First, these benefits in Brazil are not subject to payroll taxes and therefore reduce total labor costs. Second, labor legislation is generally more flexible regarding the provision of benefits such that it is easier to adjust benefits than wages ([Arbache, 1995](#)). Even though regulations for fringe benefits provision are considered less rigid

¹³The minimum wage is above 70% of the median wage in Brazil.

than for wages, collective bargaining agreements (CBA) sometimes include clauses pertaining to these benefits. In particular, the third most common clause type among extended firm-level CBA includes wage supplements such as food subsidy (Lagos, 2020). Also, around 10% of all formal sector firms are under CBA with a clause on health plan/insurance (Marinho, 2020).

But how important should be adjustments in the non-wage compensation margin in Brazil compared to other settings? Non-wage benefits should play a larger role in countries where minimum wages are high. The minimum wage as a fraction of the median wage ranges is lower in the United States (32%), Japan (44%), and Mexico (46%), than in Brazil (75%) and many European countries, for example, the UK (55%) and France (61%).¹⁴

Although transport subsidy is a mandated benefit in Brazil since 1985, we treat this as a benefit that firms can adjust. This is likely the case since we observe that only 36% of formal sector workers report they receive this benefit. That is, firms may not fully comply with all aspects of labor regulations. Also, as transport benefit is non-wage compensation, firms do not incur payroll taxes. In addition, firms may deduct the cost with the offered subsidy from the base for income taxation as well as from their operational cost, lowering net revenues which are the base for other corporate and payroll taxes.¹⁵ This implies that firms have incentives to offer transport benefits and a further incentive to adjust it at the intensive margin by providing better means of transportation or increasing the benefit in cash.

3 A Simple Theory

In this section, we describe a simple model assuming perfectly competitive labor markets to guide our analysis. We assume migrants and natives to be perfect substitutes and investigate the consequences of a migration shock that shifts the aggregate labor supply to the right. Then we introduce intersectoral linkages where formal and informal workers are substitutes.

In a standard competitive market model, where the wage is determined such as labor supply equals labor demand, migration negatively affects wages. In the extreme case in which supply is inelastic, migration has no effect on the employment of natives and all migrant workforce is absorbed. On the other hand, with an upward-sloping labor supply, the decline in wages makes jobs less attractive for some native workers such that native employment falls.

Of course, this benchmark is a vast simplification of how the labor market works in practice. Downward wage rigidities are often present in reality due to minimum wage laws and collective bargaining agreements. In this case, migration shocks can be accommodated by job losses or lower labor costs, for example, reducing non-wage

¹⁴Brazil (Source: PNAD Continua 2015). Data for OECD countries in 2019 (Source: OECD.Stat, <https://stats.oecd.org/Index.aspx?DataSetCode=MIN2AVE>)

¹⁵The income tax due cannot be reduced by more than 10%.

benefits (McKenzie, 1980; Clemens, 2021).

Also, we have considered that the formal and informal sectors are independent, which masks important intersectoral linkages, in particular, on the production side (see Ulyssea (2010, 2018) and Bosch and Esteban-Pretel (2012)). When the two types of labor are highly substitutes, informal employment and wages can compensate for wage rigidities in the formal sector. This should be particularly relevant to understanding the implications of an increase in migration in the labor market where formal wages are rigid thereby increasing the importance of the informal sector as an outside option.

To address such context, we develop a simple model with informality in which the formal sector is subject to the minimum wage and offers non-wage benefits that are frequently observed in the data (e.g. health insurance and food subsidy). Our starting point is the seminal contribution of Harris and Todaro (1970). We follow their model, in which minimum wage is the main institution behind the existence of a two-sector economy. We add non-wage benefits in the formal sector as a source of adjustment of total compensation in the presence of minimum wages.¹⁶ Appendix A presents the model and here we summarize its main predictions.

In this model, the effects of migration may depend on the degree of substitution between formal and informal labor inputs in production and on the non-wage benefits margin. Considering that migration exogenously shifts the supply of workers to the informal and formal sectors at the destination, our model has clear predictions regarding the direction of effects of migration on employment by sector, unemployment, formal sector non-wage benefits, and informal wages.

Table A1 simulates the impact of a migration shock when non-wage benefits are flexible (column 2) or fixed (column 3). It does so by assuming that production linkages are low, medium, and high, in Panels A, B, and C, respectively. In the benchmark economy with medium production linkages across sectors and flexible non-wage benefits, migration increases unemployment and informal employment but decreases non-wage benefits and informal sector wages. Formal employment remains unchanged. With non-wage benefits fixed at baseline levels, our main results show that migration has now a negative impact on formal employment as expected since formal firms cannot adjust benefits after the supply shock from migration. Consequently, unemployment and the informal sector adjust more. Under lower intersectoral linkages, the qualitative results are the same with stronger effects in informal markets and formal sector benefits.

In sum, in a simple competitive model with no rigidities and absent non-wage benefits, which is likely closer to the informal (unregulated) sector case, we expect negative effects of migration on wages and possibly on the employment of natives depending on the labor supply elasticity. As we consider that formal and informal

¹⁶We abstract from other sources of labor market frictions, which are explored in much recent work on models of the labor market with monopsony to study immigration effects (e.g. Amior and Manning (2020) and Amior and Stuhler (2022)). These are not needed to understand the mechanisms we emphasize, so we proceed with a simpler approach accounting for unemployment, two employment sectors, and intersectoral linkages.

labor inputs can be substitutes in production, minimum wage regulation, and the possibility of adjustment in non-wage benefits in the formal sector, we find that the model yields predictions that are largely consistent with a decline in formal employment of natives and in benefits offered in this sector. This reduction would be even larger if no adjustment in non-wage benefits were possible. On the other hand, as the informal sector absorbs a larger fraction of migrants, wages fall.

Empirically, these forces are likely to affect mainly those workers at the bottom and medium of the formal wage distribution since migrants are generally low skilled.¹⁷ Low-skill workers receive disproportionately more generous benefits (e.g. health insurance and food vouchers) but have wages that are more prone to be affected by minimum wage policies and collective bargaining agreements. Even if many firms employ both high and low-skilled workforce which may produce spillover effects due to complementarities, the impact on high-skilled workers is likely to be of second order.

4 Data and Empirical Strategy

In this section, we begin by listing the main sources of data used in our analysis and showing some descriptive statistics. Then we describe the empirical framework and report first-stage estimates that link observed migration patterns to our predicted migration flows.

Migration We draw data from three waves of the Brazilian Census (1991, 2000 and 2010), provided by the *Instituto Brasileiro de Geografia e Estatística (IBGE)*, to construct two of the main variables used in our study.¹⁸ First, we leverage Census answers about the municipality of origin and year of migration to construct a measure of yearly migration outflow from each municipality in the Semiarid and a measure of inflow to each destination (all but Semiarid) during the 1996-2010 period. Second, we use the 1991 Census to build a “past settlement” measure by associating the share of migrants from each Semiarid municipality who resides in each destination. In Appendix B we provide more details on how we structure our yearly migration dataset.

Weather shocks Weather data were retrieved from the Climatic Research Unit at the University of East Anglia ([Harris et al., 2020](#)). The CRU Time Series provides worldwide monthly gridded data of precipitation and temperature, at the $0.5^\circ \times 0.5^\circ$ level (0.5° is around 56km on the equator). We construct municipality-level monthly precipitation and temperature measures based on grid-level raw data as the weighted average of

¹⁷This follows the arguments developed by the labor market model in Card and Lemieux (2001) and Borjas (2003). In Section 4, we present descriptive evidence that supports greater substitutability between migrants and less skilled natives in the labor market.

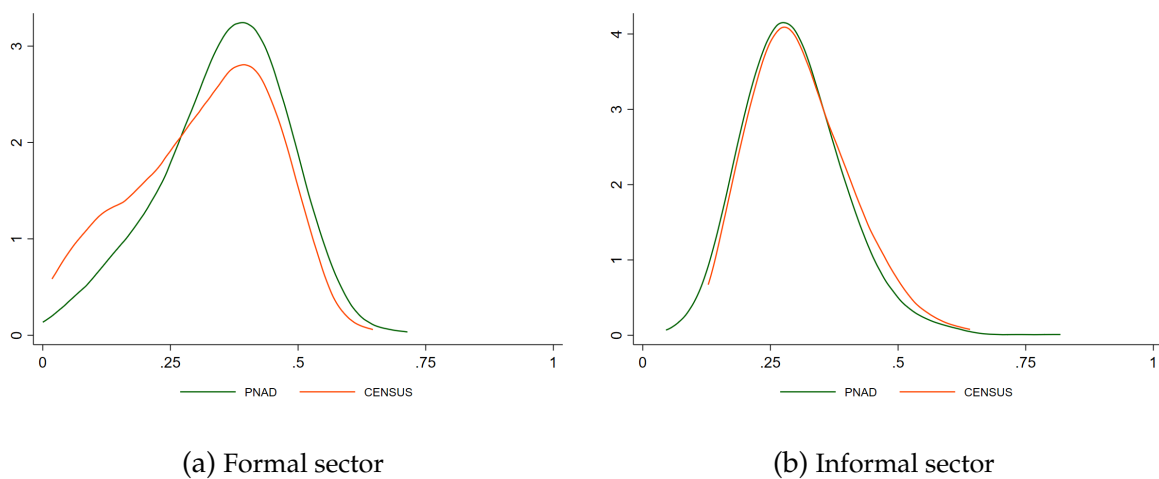
¹⁸As several municipalities were split into new ones during the 1990s, we aggregate our data using the original municipal boundaries as they were in 1991 (so-called “minimum comparable areas” or MCA) in order to avoid potential miscoding regarding migration status or municipality of origin. We use municipality and MCA as synonyms throughout the paper.

the municipality grid's four nodes using the inverse of the distance to the centroid as weights.¹⁹ We define rainfall shocks as deviations from the historical average.²⁰

Labor outcomes Our primary source of data for the outcome variables is the *Pesquisa Nacional por Amostra de Domicílios - PNAD*, a household survey also collected by the IBGE, the bureau responsible for conducting the Census. The survey is conducted yearly, except during Census years. Thus our data spans from 1996 to 1999 and from 2001 until 2009. Interviews take place in 808 municipalities in all 27 Brazilian states and cover several dimensions such as education, labor, income, fertility, and household infrastructure. On average, around 300,000 people are interviewed in each round.

Two features of *PNAD* are worth pondering regarding the suitability of the data for our analysis. First, less than 20% of all municipalities are comprised in the survey. However, they represent nearly 80% of the destination choice of migrants from the Semiarid region and 65% of the employed population in Brazil. Second, *PNAD* is not designed to be representative at the municipality level. To reassure that our primary outcome measures from *PNAD* are valid, we compare data from *PNAD* 2009 with Census 2010. In Figure 1 we compare the distribution of municipality-level employment rate in the formal and informal sectors from the 2009 round of *PNAD* with the distribution for the same *PNAD* municipalities obtained from the 2010 Census. In Figure 2 we show the log earning distributions from *PNAD* and Census in the same years. Both measures across formal and informal labor markets overlap very significantly. Unfortunately, the Census does not provide any information on non-wage benefits.

Figure 1: Distribution of employment rates, PNAD vs. Census



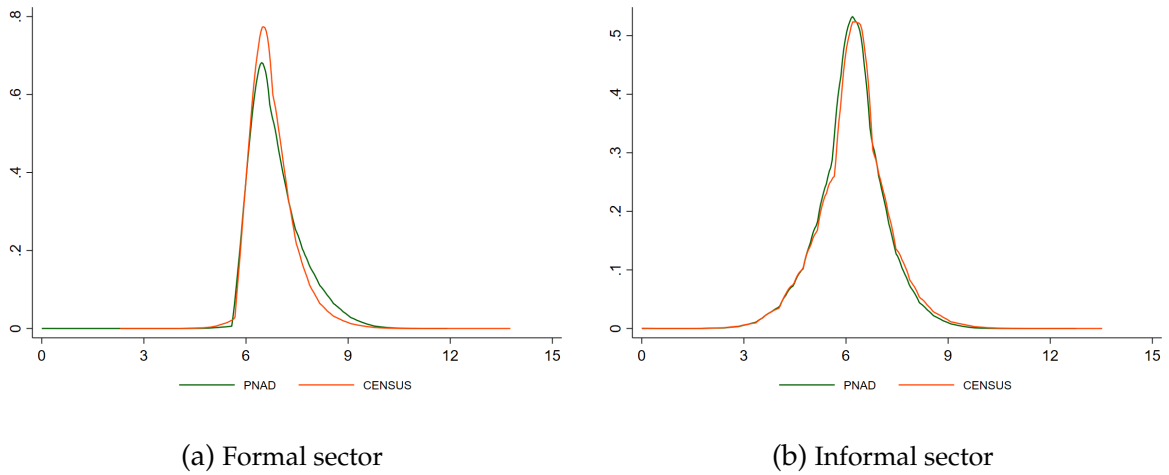
Notes: Municipality-level employment rate in the formal and informal sector from PNAD (2009) and Census (2010).

Our main outcomes comprise earnings and employment. We explore whether the worker is an employee in the registered formal sector, informal sector or self-employed;

¹⁹This approach is similar to the one used by Rocha and Soares (2015).

²⁰See Appendix C for a detailed description and discussion on this measure.

Figure 2: Distribution of log earnings, PNAD vs. Census



Notes: Log earnings in the formal and informal sector from PNAD (2009) and Census (2010).

and whether she is unemployed or out of the labor force. We also use information on non-wage compensation. The survey asks specifically whether the individual received any kind of payment or help to cover expenses with food, and transport and if the employer provides health insurance. Unfortunately data on the intensive margin of nonwage compensation is unavailable.²¹

We restricted our attention to individuals between 18 and 65 years old, living in the municipality for 10 years or more and we refer to them as *natives*. We consider as destinations all the PNAD municipalities that are not in the Semiarid in order to minimize concerns about spatial correlation in weather shocks. We pool the 13 years of individual survey data and take averages at the municipality-year level. The final destination sample has 684 unique municipalities and 8,190 municipality-year observations, averaged from 2,152,950 individuals.

Table 1 describes municipality-level data for origin (Panel A) and destination (Panel B) municipalities. Semiarid's areas show lower levels of rainfall, and slightly higher temperatures and are less populated than destination municipalities. On average, 1.0 p.p. of Semiarid's population leaves every year, resulting in an average increase of 0.30 p.p. of the labor force in the destination.

Table 2 provides descriptive statistics for destination municipalities. In our sample, 63% of individuals are employed - with 31% having a formal job, the same proportion of informal workers. The unemployment rate is 13% and 24% of individuals are not in the labor force. The average monthly earning is R\$ 637.89, with the formal sector having a substantially higher average (R\$ 788.22) than the informal sector (R\$ 491.28).²² Among workers employed in the formal sector, 39% receive financial help to cover expenses

²¹In section 5.2 we provide a back-of-the-envelope calculation of the impacts of internal migration on total compensation for native workers, using the average cost of food subsidies and private health insurance.

²²Earnings are measured in R\$ (2012).

with food, 36% for transport and 21% for health expenditures.²³

Finally, Table 3 compares migrants to low- and high-education natives. Migrants are slightly more educated and earn slightly less than less-educated natives. They also have a similar likelihood of working part-time and being in the formal sector when compared to low-education natives. On the other hand, high-education natives are more likely to work in the formal sector and have considerably higher pay. Appendix Table B1 shows that top occupations for migrants (e.g. typically bricklayer for men, domestic worker for women) are also top occupations for low-education natives, but not for the skilled. Also, the same five industries that concentrate over 80% of working migrants also employ a similar share of low-education workers (see Appendix Table B2). Overall, this characterization is consistent with greater substitutability between migrants and less skilled natives in the labor market.

4.1 Empirical Strategy

Here we first describe the empirical framework that allows us to (i) isolate the observed variation in migration induced by exogenous weather shocks, and (ii) the migration flows into destination municipalities determined by past settlements. Next, we discuss and present supportive evidence on the validity of this shift-share instrument approach based on insights from the recent econometric literature that analyzes its formal structure.

We specify a model for the changes in labor market outcomes of native individuals as a function of internal migration flows. Specifically, we assume that

$$\Delta y_{dt} = \alpha + \beta m_{dt} + \gamma X_d + \psi_t + \epsilon_{dt} \quad (1)$$

where y_{dt} is a vector of labor outcomes at destination municipality d in year t , m_{dt} is the destination migrant inflow from the Semiarid region, X_d are destination-level controls interacted with time dummies, ψ_t absorb time fixed effects and ϵ_{dt} is the error term. The main challenge to identifying β is that the observed migration, m_{dt} , is the equilibrium between the demand and supply of migrants. Another issue is that the error term, ϵ_{dt} , may include unobserved characteristics that could be correlated with migration inflows. In particular, migrants could choose a specific destination municipality due to demand shocks leading to higher wages or job prospects. By differencing the outcome variables we can account for time-invariant unobserved characteristics that could be correlated with migrant inflows, but not the time-varying confounders which would potentially bias OLS estimates.

We account for this endogeneity problem following a two-step procedure to construct an instrumental variable for the number of migrants entering a destination. First, we predict m_{ot} , the migration outflow rate²⁴ from origin municipality o in year t , using

²³Less than 1% of informal and self-employed workers receive any kind of non-wage compensation.

²⁴Defined as the observed number of migrants leaving the municipality divided by the population in the 1991 Census.

weather shocks in the previous year:

$$m_{ot} = \alpha + \beta' Z_{ot-1} + \phi_o + \delta_t + \varepsilon_{ot} \quad (2)$$

where Z is a vector of rainfall and temperature shocks at the origin municipality o in the previous year, ϕ_o and δ_t are municipality and year fixed effects, respectively, and ε_{ot} is a random error term. For each year the predicted number of migrants who leave their hometowns is obtained by multiplying this predicted rate by the municipality population reported in the 1991 Census:

$$\widehat{M}_{ot} = \widehat{m}_{ot} \times P_o \quad (3)$$

In the second step, we use the past settlements of migrants from the origin o to municipality d in order to distribute them throughout the destination areas, defining our shift-share instrumental variable (SSIV) as

$$\widehat{m}_{dt} = \sum_{o=1}^O \frac{s_{od} \times \widehat{M}_{ot}}{P_d} \quad (4)$$

where s_{od} is the share of migrants from origin municipality o who lived in the destination area d in 1991²⁵ and P_d is the total population at d in 1991.²⁶ Thus our instrument $\widehat{m}_{dt}$ can be thought of as a combination of exogenous shocks or ‘shifts’ $\widehat{M}_{ot}$ (weather-driven outflows) and exposure ‘shares’ ($s_{od} \geq 0$) or past settlement patterns.²⁷

The validity of the shift-share instrument approach relies on assumptions about the shocks, exposure shares, or both, as discussed by recent literature which analyzes its formal structure. [Goldsmith-Pinkham et al. \(2020\)](#) demonstrate that a sufficient condition for consistency of the estimator is the strict exogeneity of the shares. Alternatively, [Borusyak et al. \(2021\)](#) show how one can instead use the exogenous variation of shocks for identification by estimating a transformed but equivalent regression - at the origin level in our setup - where shocks are used directly as an instrument.

Based on these insights, we leverage origin-level weather shocks²⁸ for identification and define the reduced-form relationship that associates labor market outcomes and the predicted migrant flow at the destination as

$$\Delta y_{dt} = \alpha + \beta \widehat{m}_{dt} + \gamma X_d + \psi_t + \epsilon_{dt} \quad (5)$$

We follow [Borusyak et al. \(2021\)](#) and calculate an origin-level weighted average version of equation 5, that uses the exposure shares s_{od} as weights, and results in the transformed reduced form relationship

²⁵We fix our past settlement measure in 1991 across the time span of our sample so as to avoid concerns about the persistence in migrant flows as discussed by [Jaeger et al. \(2018\)](#). We also experimented with a specification that updates the past settlement using the data from the immediate previous Census and the results are similar.

²⁶Note that the denominator P_d is only a normalization that helps interpret the coefficients of interest. It does not play any role in identification.

²⁷In appendix C we further discuss our shift-share instrument in more detail.

²⁸Figure 12 illustrates the variation in weather shocks that use for identification.

$$\bar{y}_{ot} = \alpha + \beta \widehat{M}_{ot} + \bar{\varepsilon}_{ot} \quad (5')$$

In Appendix D we provide a detailed derivation of the transformation performed. As discussed shown in [Borusyak et al. \(2021\)](#), the consistency of our shift-share approach is based on two conditions:

Assumption 1 (*Quasi-random shock assignment*): $\mathbb{E}[Z_{\tilde{o}}|\bar{e}, s] = \mu$ for all $\tilde{o}$.

Assumption 2 (*Many uncorrelated shocks*): $\mathbb{E}[\sum_o s_o^2] \rightarrow 0$ and $Cov[Z_{\tilde{o}}, Z_{\tilde{o}'}|\bar{e}, s] = 0$ for all $\tilde{o}, \tilde{o}'$.

where $\tilde{o} = (o, t)$, $\bar{e} = \{\bar{e}_{\tilde{o}}\}_{\tilde{o}}$, $s_o = \sum_d s_{od}$ and $s = \{s_o\}_o$.²⁹ Assumption 1 guarantees that our shift-share IV is valid when weather shocks are as-good-as-randomly assigned, which comes from standard natural shocks arguments. Given identification, Assumption 2 gives us consistency when the number of observed shocks is large and when shocks are mutually uncorrelated given the unobservables and s_o . In the Appendix Table D1 we show that the effective sample size³⁰ is sufficiently large, reassuring us that exposure concentration is not a relevant issue in our setting. Also, in Appendix Tables E2-E5 we present evidence that the shocks we are using can be treated as uncorrelated, which supports the validity of Assumption 2 in our setting.

One additional advantage of using the origin-level shocks concerns hypothesis testing. [Adao et al. \(2019\)](#) show that conventional inference in shift-share regressions is generally invalid because observations with similar exposure shares are likely to have correlated residuals, potentially leading to null hypothesis overrejection. [Borusyak et al. \(2021\)](#) show that by using the shock-level relationship instead of the destination-level one can obtain standard errors that converge to those obtained by the [Adao et al. \(2019\)](#)'s correction procedure.

4.2 Weather-induced Migration

We begin the exploration of our first-stage results by estimating variations of specification 2 and report the estimates in Table 4. All regressions control for temperature shocks and the log of the total population in the previous census; and include time and municipality fixed effects. In columns (2)-(8) we include a flexible trend interacting time dummies with 1991 characteristics (age and the shares of high school and college-educated individuals). Columns (3)-(6) include up to three lags, contemporaneous and one lead of rainfall and temperature shocks. For brevity, we omit (mostly insignificant) coefficients associated with temperature shocks in Table 4. Standard errors are clustered

²⁹As in [Borusyak et al. \(2021\)](#), $\bar{\varepsilon}_{ot}$ correspond to the error term from equation 1 computed at the level of shocks (e.g. municipality of origin).

³⁰According to [Borusyak et al. \(2021\)](#), the effective sample size is the inverse of the Herfindahl index of concentration of migrants: $H = \frac{1}{\sum_o s_o^2}$.

at the grid level to account for the fact that municipalities in the same grid will have similar shocks.³¹

As expected, rainfall shocks in the previous year are negatively correlated with migration outflows indicating that Semiarid’s inhabitants leave the region during drought periods. Coefficient estimates are remarkably stable across specifications and adding more lags does not change the baseline results. More important to our identification, we include as control rainfall and temperature shocks one year forward to ensure that our instrument is not contaminated by serial correlation in the weather measures. The coefficient on $rainfall_{t+1}$ reported in column (6) is small in magnitude and not statistically significant, while the coefficient for $rainfall_{t-1}$ remains almost unchanged. Our estimates indicate that a municipality where annual rainfall is 10% below the historical average will experience an increase of 1p.p. in migration outflow rate.

Next, we distribute the predicted migration outflow shock using past settlement patterns of migrants from the origin municipality o to destination d . A *sine qua non* requirement implicit in our empirical framework is that both predicted migration outflow and inflow rates, $\hat{m}_{ot}$ and $\hat{m}_{dt}$ respectively, should be strongly correlated with their observed counterparts. Figure 3 illustrates that our predictions provide a strong fit of the observed migration. Panel (a) shows the relationship between the predicted and observed number of migrants leaving the Semiarid region and entering non-Semiarid municipalities, accumulated over the period 1996-2010. Panel (b) shows the predicted and observed numbers of incoming Semiarid migrants for destination municipalities.

In Appendix C we describe in more detail our data source for weather shocks, discuss alternative measures of weather, and present further details about how we constructed our instrument including predicted and past settlement patterns.

Overall, this analysis shows that our strategy provides a strong first-stage as predicted migration rates, $\hat{m}_{dt}$, are strongly correlated with observed migration. Appendix Table D1 reveals that our first-stage point estimates are close to a one-to-one relationship (0.92) - making the magnitude of reduced-form and IV estimates almost identical - and have an F-stat of 2,275.³²

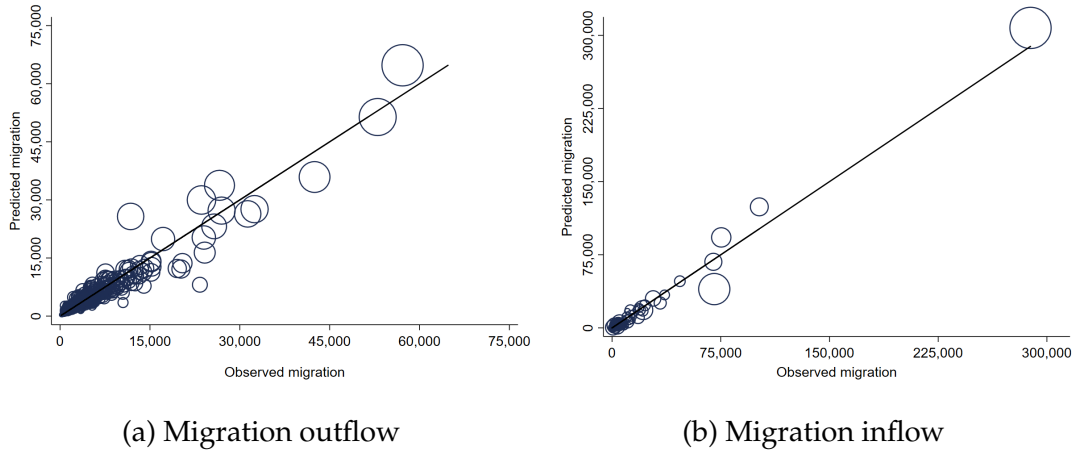
5 Labor Market Effects of Migration Inflows

Now we turn our attention to labor markets at the destination and investigate how migration inflows affect earnings, employment, unemployment, and labor force participation of native workers. Next, we explore how labor markets adjust to migration shocks in terms of non-wage compensation.

³¹Similar, but not identical, as shocks are computed by taking the average of the grid’s four nodes, weighted by the inverse of the distance from each node to the municipality centroid. Therefore, two municipalities inside the same grid have different shocks because the distance to the centroid is not the same.

³²A sufficiently high F-stat avoids weak instrument concerns, especially in the light of the recent discussion in Lee et al. (2020) who show that a 5 percent test requires an F statistic of 104.7, significantly higher than the broadly accepted threshold of 10.

Figure 3: Observed vs predicted migration



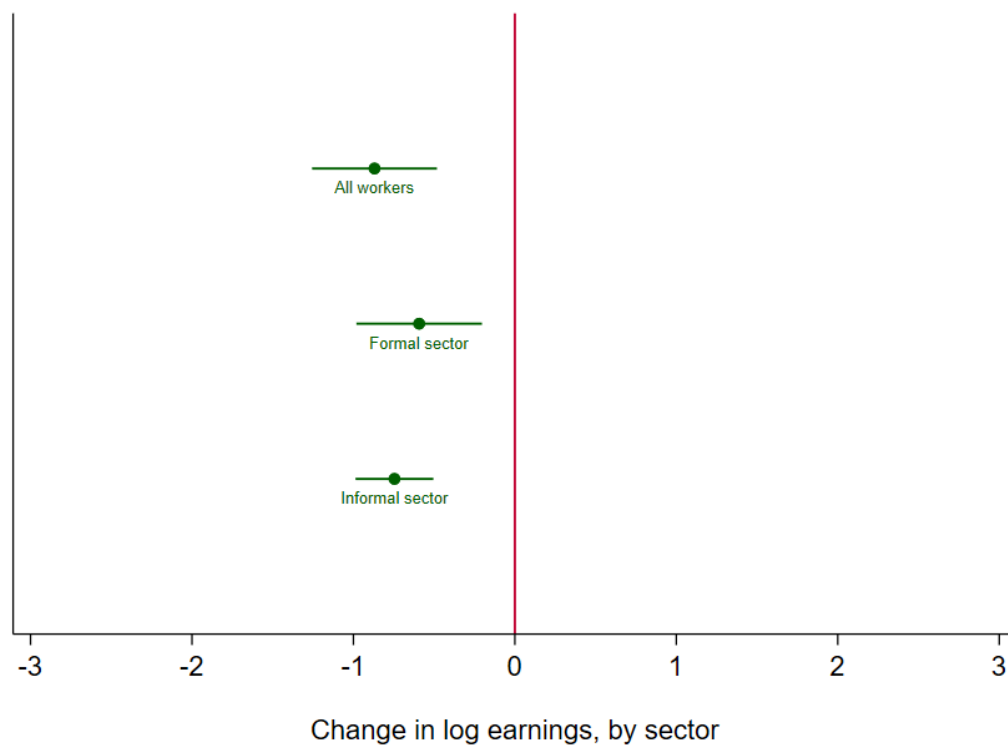
Notes: This figure presents the relationship between the predicted and observed migration flows across Brazilian municipalities from 1996 to 2010. Panel (a) shows the number of migrants leaving the Semiarid region to non-Semiarid municipalities. Panel (b) shows the number of incoming Semiarid migrants for destination municipalities. The circle size represents the municipality's total population in 1991. *Data source:* Census microdata (IBGE).

We begin by investigating how native workers' earnings adjust to exogenous migration inflows. Table 5 reports several specifications for our SSIV estimates. Column (1) displays a flexible specification, without any control. In column (2) we include time dummies and in column (3) we also control for a vector of destination-level characteristics measured in 1991 (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. All regressions are weighted by the working-age native population in 1991. Standard errors are clustered at the origin municipality level.

Panel A reveals a negative effect of the inflow of Semiarid's migrants on average log earnings for native workers. Adding covariates lowers somewhat the magnitude of our estimates but does not change substantially our main conclusions. One percentage point increase in the number of migrants reduces earnings by 0.87%. In Panel B we restricted our analysis to native workers holding a formal job, while in Panels C we focus on those in the informal sector, including workers who are self-employed. We find that a one percentage point increase in the inflow of migrants reduces the earnings of formal workers by 0.59% and by 0.75% for those employed in the informal sector.

Figure 4 summarizes our main findings. Any downward wage restrictions, such as minimum wages or collective bargaining agreements, may alleviate the impacts of the incoming migration on earnings for natives employed in the formal sector. The negative effect on informal earnings is greater, albeit only slightly. That is consistent with the absence of downward wage rigidity in this sector such that the classic predictions from perfect competition prevail.

Figure 4: Effects of internal migration on earnings

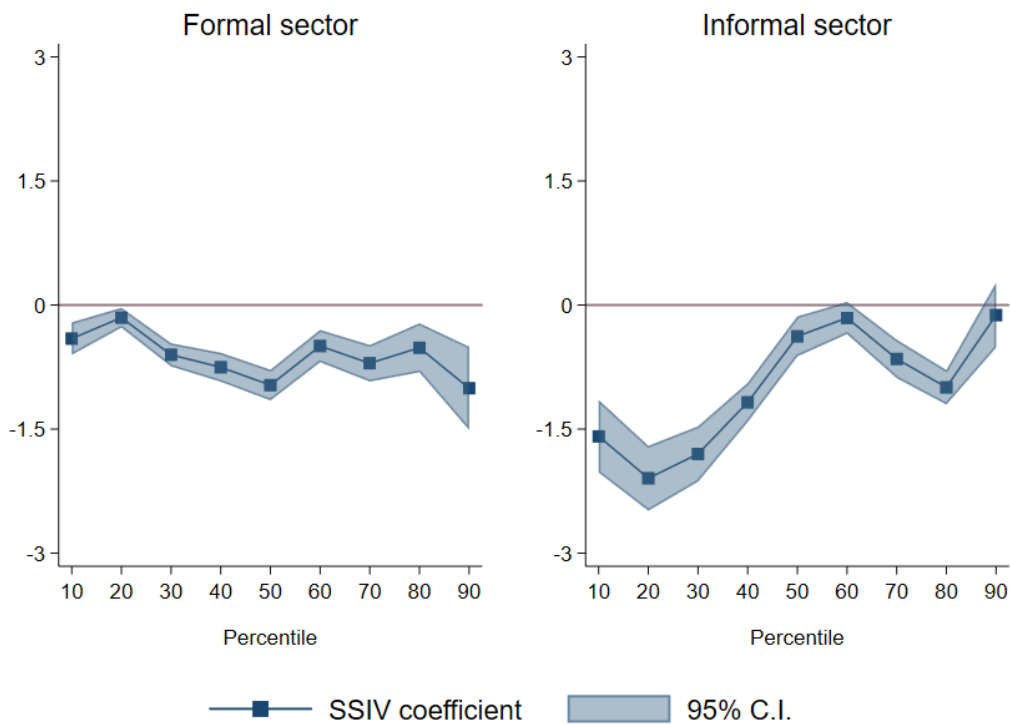


Notes: This figure plots origin-level SSIV coefficients on change in log earnings, by sector. The informal sector includes self-employed workers. Controls include time dummies and destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. Green markers are statistically significant at the 5% level.

The magnitude of our estimates is larger than that found by [Imbert et al. \(2022\)](#) (0.15%). An important difference between our approaches is that their estimates come from changes over a six years period, while we use year-on-year changes. On the other hand, our estimates are lower than those from [Kleemans and Magruder \(2018\)](#) (0.97%). In their paper, they estimate the effects on the log income per hour - so they may also be capturing the adjustment on hours worked - while we focus on log monthly earning. Similarly, they also find a larger effect on the income of informal workers compared to those in the formal sector.

We also investigate the differential effects according to the native worker's position in the earnings distribution. We calculate the municipality-level thresholds of each decile of the earnings distribution, separately for each sector, and run a regression on the changes of each threshold. Figure 5 reports our results. For those workers employed in the formal sector, we find smaller impacts at the bottom of the distribution. This is consistent with wage rigidity in the formal sector which limits the negative impacts for low-paid workers. For informal workers, the impact is substantially stronger for those at the bottom third of the distribution, consistent with classic predictions from perfect competition and greater substitutability between migrants and less skilled natives

Figure 5: Effects of predicted migration along the earnings distribution



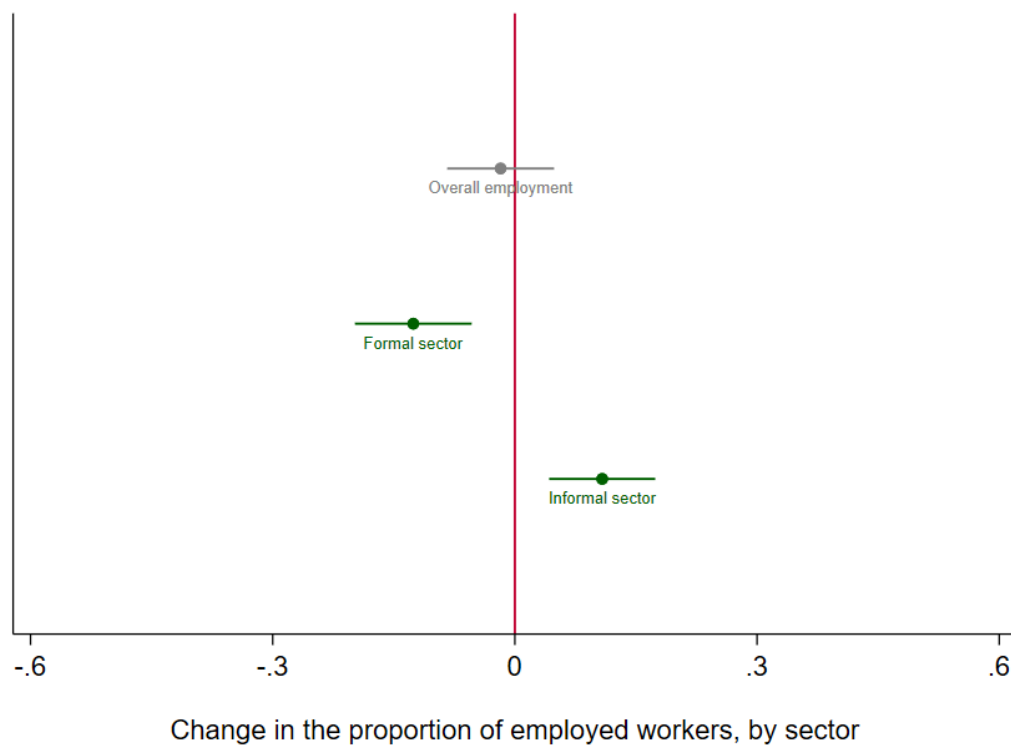
Notes: This figure plots SSIV coefficients of change in the average of log earnings, in each decile, by sector. The informal sector also includes self-employed workers. Controls include time dummies and destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies.

in this sector. To a smaller extent, migration also affects higher earnings deciles of informal sector workers and self-employed. The negative impact of migration, in this case, may be attenuated due to some formal sector workers moving into informality or self-employment. As workers in the formal sector are more productive, on average, this increases earnings at higher percentiles in other sectors.

Our results for employment are summarized in Figure 6. While we find no effect on overall employment, the inflow of migrants from the Semiarid does change the composition of workers across sectors. Table 6 reports the point estimates across all specifications. Our estimates in Panels B and C imply that a one percentage point increase in the inflow reduces the share of formal employment by 0.13*p.p.*, and increases the share of informal by almost the same amount (0.11*p.p.*). This results is largely consistent with the predicted fall (increase) in formal (informal) employment following the two-sector labor market model presented in Section 3. We argue that the existence of medium production linkages across formal and informal sectors, and some degree of rigidity in the adjustment of non-wage benefits can account for these findings.

Kleemans and Magruder (2018) find a larger negative effect on formal employment (-0.33*p.p.*) and no significant impact on the informal sector. In contrast, we find evidence

Figure 6: Effects of internal migration on employment



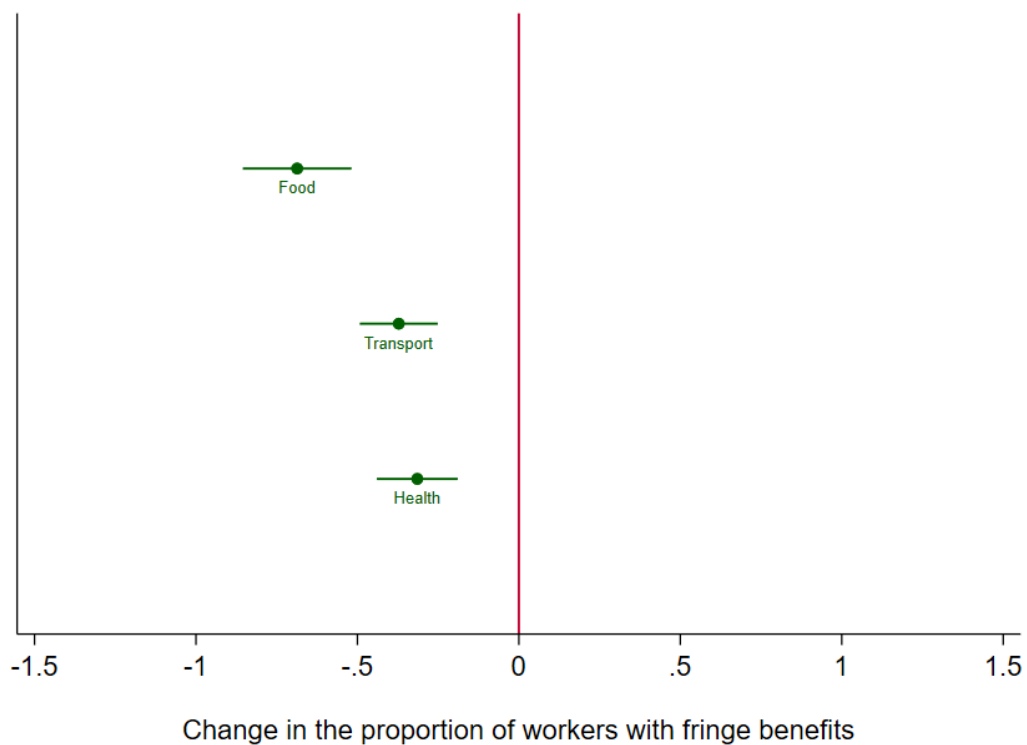
Notes: This figure plots origin-level SSIV coefficients of change in employment rate, by sector, measured as a fraction of the native working-age population in 1991. The informal sector includes self-employed workers. Controls include time dummies and destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. Green markers are statistically significant at the 5% level.

consistent with the reallocation of workers between sectors. This is likely because migrants are mostly high-educated individuals in their setting, while in our case, they are more likely to be low-educated and, therefore, more likely to compete with low-educated native workers.

To draw a more complete picture we also estimate the impacts on unemployment and labor force participation reported in Table 7. Migration inflows lead to an increase of 0.09*p.p.* in the unemployment rate and a decrease of 0.08*p.p.* in the proportion of out-of-labor-force individuals. What mechanism accounts for these estimates is ex-ante unclear. On one hand, increased competition in the labor market could discourage native individuals to work if wages or benefits fall, as predicted by the model developed in the Appendix A. On the other hand, if the primary earner in the household loses his/her job because of the increased competition, then it is possible that other members of the household would enter the market, a phenomenon known as the added worker effect (Lundberg, 1985).

We test this second mechanism by running the same regressions separately for individuals identified as head or non-head of the household. According to Table 8, almost all of the employment effects come from the head of households, while the

Figure 7: Effects of internal migration on non-wage compensation



Notes: This figure plots SSIV coefficients on change in the proportions of formal sector workers who receive health insurance, food, or transport subsidies. Controls include time dummies and destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. Green markers are statistically significant at the 5% level.

changes in unemployment and inactivity rates stem from non-head members. That suggests that the second channel prevails. Also, the symmetry between the effects on unemployment and inactivity indicates that once secondary earners enter the market, it takes time for them to find a job.

5.1 Non-wage Compensation.

We now explore an additional margin of adjustment due to migration shocks. As firms operating in the formal sector cannot reduce wages below the legal minimum, they may adjust to labor supply shocks by reducing fringe benefits as discussed in Section 2.2. We focus on individuals who are currently holding a formal job because these benefits are almost exclusively offered by formal firms. Figure 7 reports the estimates. A one percentage point increase in the predicted number of migrants reduces the share of workers receiving food subsidy between 0.33p.p. and 0.69p.p., transport between 0.37p.p. to 0.52p.p., and health insurance in the range of 0.31p.p. to 0.47p.p.. See Table 9 for the underlying point estimates.

Next we complement these estimates by focusing on the behavior of firms as

providers of health insurance to their employees.³³ Instead of relying on survey data, here we turn to firm-level administrative data on health insurance contracts obtained from *Agência Nacional de Saúde Suplementar (ANS)*, the Brazilian regulatory agency responsible for overseeing the private health industry. They provide information about every employer-sponsored contract signed going back as far as 1940. We have data on the date when the contract was signed and the firm unique identifier, which we can use to merge with *RAIS*, an employer-employee matched dataset obtained from the Ministry of Labor, that provides firm-level data on the near universe of formal employment contracts. We define an indicator variable $y_{idt} = \mathbb{1}(t \geq t^s)$ for each firm i in the destination municipality d at year t , with t^s being the year when the health insurance is hired. Then we estimate how migration inflow rates at destination municipality d affect changes in y_{idt} , that is, the likelihood that firm i provides health insurance to its employees.

In column 1 of Table 10 we find that firms operating in a municipality that receives more incoming migrants are on average less likely to provide health insurance to employees.³⁴ An increase of one standard deviation in migration rate of 1p.p. reported in Table 1 implies a 1.5p.p. decrease in the share of firms that provide health insurance, roughly the average of y_{idt} . In Columns 2-5 we restrict the sample to different bins of firm size. The effect is close to zero and insignificant for firms below 100 employees, but negative and of greater magnitude for larger firms. Firms above 100 employees are at least 6 times more likely to provide health insurance as part of compensation.

5.2 Total Compensation.

To address whether changes in non-wage compensation are important relative to the changes in earnings – and which types of compensation are more important – we would need knowledge of the willingness to pay measures for each benefit in such context. One way to calculate the change in non-wage compensation in dollar equivalents is to attach a dollar value to each type of non-wage compensation. For example, one could value food subsidy as the firm’s average subsidy cost. Under this assumption, i.e., that individuals value each benefit at its cost ($value = \text{average wage cost}$), we estimate the variation in total compensation, ΔTC_{dt} , as follows:

$$\Delta TC_{dt} = \Delta W_{dt} + \sum_i value_i * \Delta B_{dt}^i \quad (6)$$

where ΔW_{dt} is the wage variation and ΔB_{dt}^i the variation in the share of benefit type i .

We use the average cost estimates provided in [Arbache and Ferreira \(2001\)](#) for food subsidy (57% of one minimum wage or 45% of average formal wage in our sample).

³³20% of workers get private health insurance through their employers. In 2018, the average employer-provided health insurance benefit cost on average R\$582, or 17% of total compensation in that same year ([ANS, 2018](#)). See section 2.2 for more details.

³⁴All the regressions are weighted by the number of employees in the firm in 1996, the first year in our sample.

We also use the cost of employer-provided health insurance from the Brazilian Federal Health Agency data ([ANS, 2018](#)) which is R\$582 in 2018 or 52% of the average formal wage in our sample. The estimated effects on formal earnings and on the fraction of non-wage benefits are obtained from column 3 in Tables 5 and 9, respectively.

This simple exercise shows that a 1p.p. increase in the predicted number of migrants reduces total compensation by 1.06% when accounting for food subsidies and private health insurance. Compared to the effect on the formal wage, which declines by 0.59%, this means that if we do not account for the change in non-wage compensation we underestimate the change in overall compensation by 80% percent, at least when workers value such benefits at their cost estimate.

5.3 Exploring the Mechanism

As our simple model in section 3 highlights, in a perfectly competitive labor market, an increase in the number of incoming migrants would be accommodated by a decrease in wages. In the presence of downward wage rigidities, the adjustment should occur, at least partially, in other margins, such as lower employment. In our empirical setting, downward rigidities are more present in the formal sector relative to the more flexible informal market due to minimum wage and collective bargaining.

As discussed in section 5, the estimated negative impact on earnings is slightly greater in the informal sector. However, the magnitude of this wedge depends on the difference in the labor supply elasticities in each sector. In places with a more binding minimum wage, the adjustment should occur mostly via formal employment and informal earnings. In regions where the minimum wage bite is lower, it works primarily through total compensation, while the impacts on employment would be limited. Finally, nonwage compensation may help introduce some wage flexibility which may, in principle, reduce job losses.

In order to provide a formal test for this mechanism, we calculate the baseline Kaitz index (minimum-to-median wage rate) in 1996 and divide the sample into two groups. Municipalities with high (low) minimum wage bite are defined as those above (below) the median of the Kaitz index. Then, we run the same specification in column 3 from Tables 5, 6, and 9 separately for each group. Tables 11 and 12 present our findings.

In regions where the minimum wage is more binding (columns 1-3), the informal sector suffers a much stronger negative effect on earnings. 1p.p. increase in migration rate decreases informal earnings by 1.6%. Regarding the employment margin, the magnitude of the increase in the informal sector is the same as the reduction in formal employment, consistent with native workers who lost their formal jobs migrating to the informal sector. Finally, although there is a slight increase in the provision of health insurance to formal workers, the impact on overall nonwage benefits is small. This is expected as regions with more-binding minimum wages also tend to have stronger collective bargaining and, thus, more rigid benefits ([Neumark and Wascher, 2006](#))

When the minimum wage is less binding (columns 4-6), we find little effect on

informal employment and earnings. Consistent with the mechanism discussed above, the adjustment in the formal sector occurs essentially in total compensation - both on earnings and non-wage benefits. Increasing the in-migration rate by 1*p.p.* reduces formal earnings by 0.72%, and the share of formal workers receiving subsidies for food, transport and health insurance by 0.85*p.p.*, 0.48*p.p.* and 0.52*p.p.*, respectively.

5.4 Sensitivity and Heterogeneity Analysis.

In this section, we summarize a series of robustness checks we have performed to assess the validity of our main findings. Then we study the heterogeneity of our main estimated effects with respect to workers' education level.

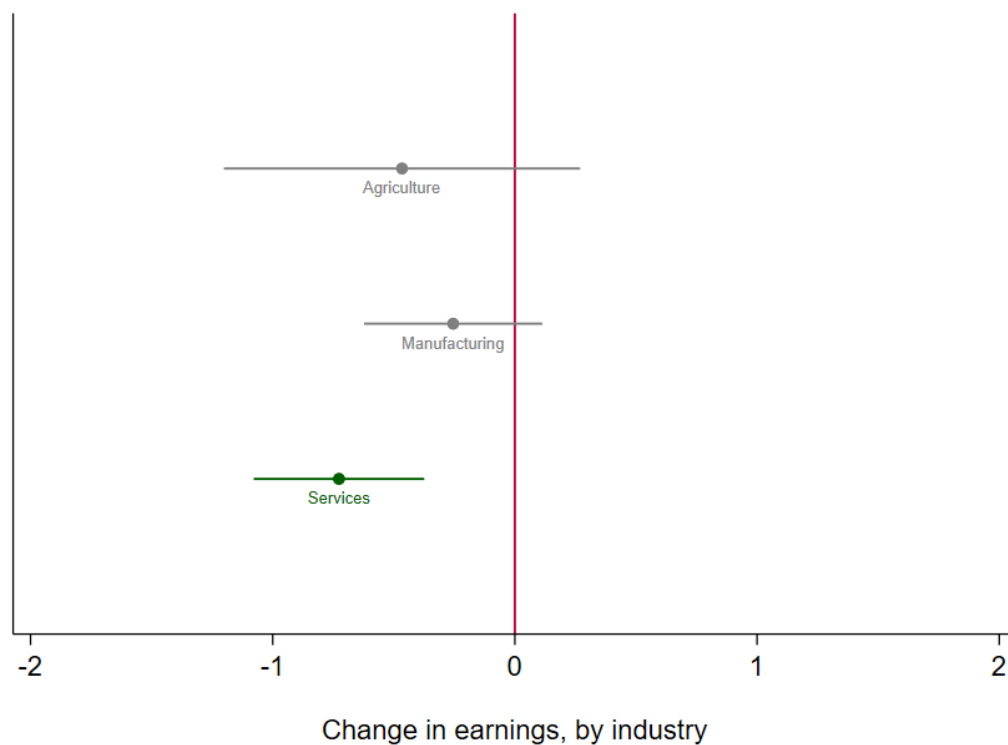
The first issue we address is whether a shift in local labor demand may be confounding our identification. If that were the case, then we should expect that migrants from other regions outside the Semiarid would be attracted to the same destinations. In other words, we should observe a positive correlation between migrant inflows from the Semiarid and that from other regions. In Table 13 we show the coefficients from a destination-level regression of the migration inflow rate of migrants from other regions on the predicted inflow rate of migrants from the Semiarid. Column (1) includes time and municipality fixed effects, while in Column (2) we add the same set of controls from our main results. Point estimates are close to zero and not statistically significant in any specification.

The second issue is that our strategy relies on the assumption that rainfall at origin municipalities affects destination labor markets only through internal migration. One possible violation of this assumption would be if a negative income shock at the origin, due to low rainfall levels, had reduced trade flows with some of the destination areas, for instance. In this case, one should expect higher effects in those industries more exposed to trade shocks, like agricultural or manufactured goods. In Figure 8 we summarize the effects from regressions of changes in log earnings on the predicted migrant inflow rate, separately by industry where the individual is employed. In Table 14 we report the detailed results. There is no statistically significant effects on the earnings for workers in the agricultural or manufacturing industries. All the impact comes from those native workers employed in services, which are less likely to be affected by negative shocks at the origin municipalities.

Finally, we explore the sensitivity of our results according to the degree of aggregation of regions of origin. In Appendix D we argue that the consistency of our shift-share instrument needs origin-level shocks to be mutually uncorrelated. As rainfall shocks are likely correlated across smaller geographical units, in Appendix E we investigate this issue by re-constructing our instrument according to larger catchment areas of origin of a migrant - such as a microregion or a mesoregion - instead of a municipality.³⁵

³⁵IBGE (1990) defines microregions as "groups of economically integrated municipalities sharing borders and structure of production". Mesoregions are collections of microregions of which not all municipalities share borders. The Semiarid has 960 municipalities, 137 micro, and 35 mesoregions.

Figure 8: Effects of internal migration on earnings



Notes: This figure plots origin-level SSIV coefficients on change in log earnings, by industry. Controls include time dummies and destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. Green markers are statistically significant at the 5% level.

First, we document that spatial correlation among shocks decreases dramatically as we consider larger areas. Second, Tables E2-E5 show that our results associating migration and rainfall, earnings, employment, and non-wage benefits remain virtually unchanged, indicating that spatial correlation among rainfall shocks in origin municipalities are irrelevant to our results.

Next, we assess whether individuals with different levels of education may experience differential impacts. In particular, we expect those low-education native workers to be close substitutes for migrants. Thus we reestimate the effect of migration on local labor market outcomes of natives with low and high education, separately. We define as less educated those with up to 8 years of schooling, which is equivalent to complete elementary education. In our sample, 58% of natives are less educated.

Figure 9 illustrates the estimates by education level. Panel A shows the effect of predicted migration on the changes in employment rates, by sector and education group. Our results are consistent with less educated native individuals being more likely to exit the formal sector and become informal sector workers compared to those who have higher levels of education. In Panel B we analyze the differential effects on log earnings. In the formal sector, there is no significant impact on native workers across

Figure 9: Effects of migration on employment and earnings, by education level

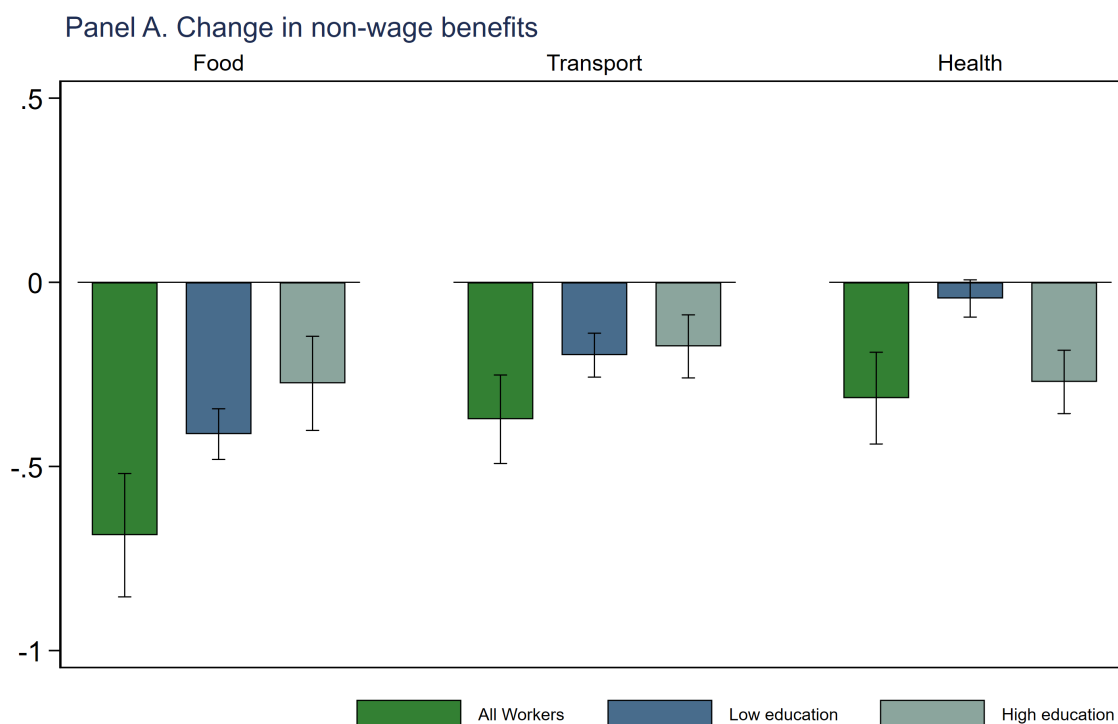


Notes: This figure plots SSIV coefficients of change in labor market outcomes, by education level. In Panel A, the dependent variables are the changes in employment rates while in Panel B we present estimates for changes in log earnings, for each sector. Each bar represents the SSIV coefficient for a separate regression on the average and by education (low education = up to 8 years of schooling). All regressions are weighted by the working-age native population in 1991, include time dummies and control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. The capped lines show the 95% confidence intervals.

education levels. This is again consistent with wage rigidity due to minimum wages or contractual wages preventing downward adjustments in the formal sector. On the other hand, native workers with low education have a relatively higher loss in informal and self-employment earnings, consistent with the conjecture that they compete more directly with (less educated) migrants.

In terms of adjustments on the non-wage benefits margin, it is less clear why they should differ by worker skills. In principle, working in the same firm implies that workers of different skills are offered a common benefits package. However, if there is a positive matching in the labor market with low (high) education workers selecting into less (more) productive and small (large) firms, then we should expect the less educated workers to be the most affected as the minimum wages bind more tightly in the firms where they work. In Figure 10 we show that the negative impact on food and transport benefits are indeed stronger and relatively more precise for low-education workers. In contrast, high-education workers have a clear reduction in employer-provided health insurance which is consistent again with some selection of these workers in large firms

Figure 10: Effects of predicted migration on non-wage benefits, by education level



Notes: This figure plots regression coefficients of change in non-wage benefits, by education level, against the predicted number of migrants from the Semiarid region in each destination municipality, measured as a fraction of the native working-age population in 1991. The dependent variables are the changes in the proportions of native workers in the formal sector who received some help to cover expenses with food, transport or health insurance. Each bar represents the reduced form coefficient by education level (low education = up to 8 years of schooling). All regressions are weighted by the working-age native population in 1991, include time dummies and control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. The capped lines show the 95% confidence intervals.

which tend to offer health insurance and where there is a mix of high and some low education workforce. A possible explanation is that the inflow of migrants competing with native low-education workers in large firms pressures wages down. However, under minimum wage restrictions, the adjustment occurs through lowering health insurance.

Changes in the benefits can have important welfare implications. We found that migration lowers the provision of food and transport benefits to less educated individuals. On the other hand, we show that health insurance is not significantly changed for low-education workers on average, while it is less offered for the high-education workers. Considering that food and transport are the two most offered benefits in the data (as shown in Table 2) and to the extent that workers value these benefits, their reduction together with a stronger negative impact on earnings for the low education workers suggest that the welfare of the less educated workers declines more than for high education workers.

5.5 Long Run Effects.

Here we turn our attention to the dynamics of the impact of migration on local labor markets in Brazil. Short and long-run effects might differ as markets adapt to current shocks. [Jaeger et al. \(2018\)](#) report short-run local effects of migration inflows for the US in the 1970s that are more negative than many in the previous literature, suggesting that the initial impact on natives is potentially large. However, they also show that much of this decline is reversed in later periods.

We account for these long-run effects by calculating the long differences in the outcome variables from 1996-2001 and 2001-2009.³⁶ We stack the two periods and estimate the same origin-level SSIV regressions from Section 5.

Table 15 shows the long-term effects of the inflow of migrants from the Semiarid region on the changes in earnings and employment. In the long-run destination labor markets adjust further, resulting in more negative impacts for the native workers. The average earnings reduce by 0.66% and 1.57% among workers in the formal and informal sectors, respectively. On the employment margin, our estimates show a decrease of .26*p.p.* in the formal sector, but no significant effect in the informal sector. Such a result may be reflecting the dual nature of formal and informal markets. In the more rigid formal sector, the markets adjust more slowly than in the flexible informal sector. Table 16 shows that non-wage benefits also are an important margin of adjustment in the long run. There is no change in the proportion of workers receiving food vouchers, but the share of natives who receives transport subsidies decreases by 0.75*p.p.* and those with health insurance reduce by 0.38*p.p.*.

A potential mechanism behind these dynamics is that short-run effects might be partially offset by further internal migration as natives respond to the adverse effects by moving to markets that were not directly targeted by migrant arrivals. Table 17 reports coefficients of the effect of our predicted shocks on the migration outflows of natives, according to levels of schooling. All estimates are positive but not very precisely estimated, and the magnitude is greater for natives of lower education, who are the most affected by the arrivals of Semiarid migrants.

6 Discussion and Concluding Remarks

In this paper, we investigate the labor market impacts of weather-induced internal migration in Brazil. We use a shift-share instrument approach combining variation in the number of people leaving their hometowns, driven by weather shocks, with past settlement patterns to exploit exogenous variation in the number of migrants entering each destination municipality.

Overall our results indicate that an exogenous supply shock of low-skill workers reduces earnings in the unregulated informal sector, especially at the bottom of the wage distribution. To a lesser extent, earnings also drop in the formal sector, with close to zero

³⁶PNAD data are not available for the years when the Census are collected - 2000 and 2010.

estimates at the bottom as minimum wage restrictions and collective agreements are more binding. Adjustments in non-wage benefits such as food vouchers, transportation subsidies, and health insurance compensate for these rigidities. From a back-of-the-envelope exercise, we find that if we do not account for the non-wage compensation margin, we tend to underestimate the change in overall compensation by 80% percent, at least when we assume that workers value benefits at their cost estimate.

We also observe a decrease in the formal employment of natives due to wage rigidities and an imperfect adjustment of the benefit margin. In the informal sector, an increase in employment follows a significant fall in earnings, consistent with workers reallocating from the formal to the informal sector or self-employment. Unemployment and labor force participation also increase, in part, due to non-head members of the households joining the job market in response to migration shocks. Most of the adjustment occurs via formal employment and informal earnings in places with a more binding minimum wage. In contrast, it works primarily through total compensation with limited employment impact in regions where the minimum wage bite is lower.

As discussed in Section 3, our model generates predictions for the impact of migration on labor markets with two sectors in an economy with different levels of intersectoral linkages and with endogenous or fixed benefits. From low to medium levels of linkages, the impact of migration in terms of employment and wage drops in magnitude. Allowing firms to adjust benefits as a response to shocks, also softens the impacts as expected. In summary, our estimates are broadly consistent with lower/medium levels of linkages and with imperfectly flexible benefits, as formal employment drops likely due to collective agreements over non-wage benefits.

Taking stock, our findings call attention to the fact that non-wage benefits are a relevant margin of adjustment for firms, especially in labor markets where minimum wages are binding and benefits are commonly offered, such as in many European and Latin American countries. They ease constraints and allow employers to absorb part of the shocks, lowering the impact on employment.

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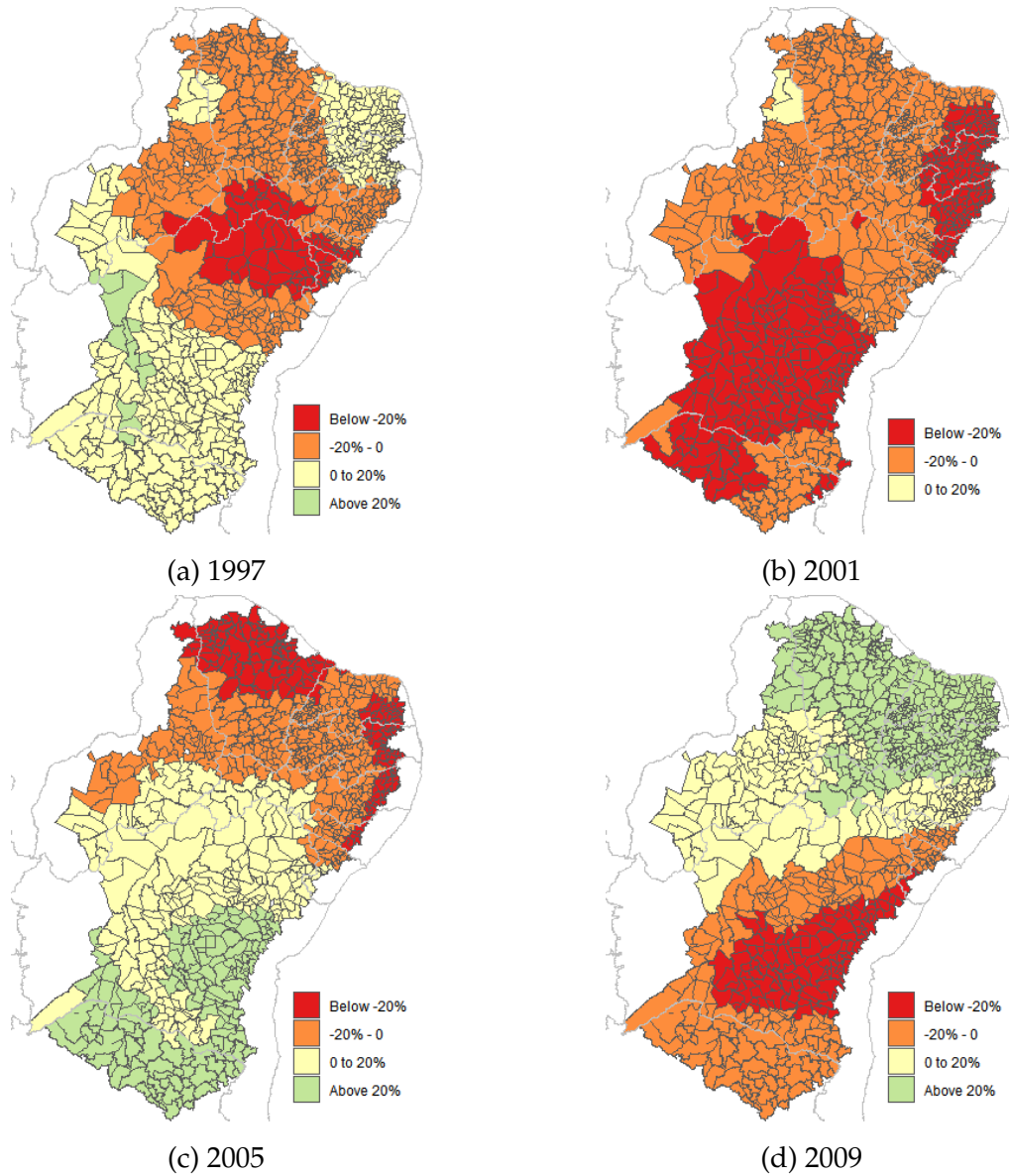
Figures and tables

Figure 11: Precipitation level: Semiarid vs Non-Semiarid



Notes: This figure compares the average precipitation level for the Semiarid region and the rest of the country, from 1996 to 2010. *Data source:* CRU Time Series v4 ([Harris et al., 2020](#)).

Figure 12: Precipitation levels in the Semiarid region for selected years



Notes: This figure presents the distribution of rainfall across the Semiarid region municipalities for selected years. Rainfall is measured as the log-deviations from historical averages. *Data source:* CRU Time Series v4 ([Harris et al., 2020](#)).

Table 1: **Summary statistics: weather and migration data**

Panel A: Origin (Semiarid)	Mean	Std. Dev.	Min	Max	Obs
Annual Rainfall	782.33	248.71	165.49	1,953.17	14,400
Rainfall shock	-0.02	0.19	-0.73	0.48	14,400
Annual Temperature	25.54	1.39	21.42	28.93	14,400
Temperature shock	0.01	0.01	-0.01	0.05	14,400
Out-migration	214.16	323.66	0.00	5,773	14,400
Out-migration rate (p.p.)	1.05	0.62	0.00	7.22	14,400
Population	21,377	30,386	1,265	480,949	14,400
Panel B: Destination (Non-Semiarid)	Mean	Std. Dev.	Min	Max	Obs
Annual Rainfall	1,610.44	401.69	660.63	3,618.55	8,190
Rainfall shock	0.04	0.16	-0.77	0.65	8,190
Annual Temperature	23.15	2.82	15.82	28.77	8,190
Temperature shock	0.03	0.02	-0.03	0.08	8,190
In-migration	146.69	896.95	0.00	25,423	8,190
In-migration rate (p.p.)	0.30	1.00	0.00	27.95	8,190
Native population	51,963	231,29	290	4,771,961	8,190

Notes: Rainfall is measured in mm. Temperature is measured in degrees Celsius. Migration outflow (inflow) rate is the share of migrants over the local (native) population.

Table 2: **Summary statistics:**
Native individuals in destination municipalities

Individual Characteristics					
	Mean	Std. Dev.	Min	Max	Obs
Female	51.08	3.65	0	72.72	8,190
Black	6.23	5.98	0	53.85	8,190
Mulatto	40.32	24.48	0	100	8,190
White	52.82	25.47	0	100	8,190
Age	37.45	1.96	30.15	55	8,190
Years of schooling	6.58	1.78	0	13.52	8,190
Less than elementary	65.33	15.75	4.71	100	8,190
Employment					
	Mean	Std. Dev.	Min	Max	Obs
Any Employment	62.72	7.95	10	100	8,190
Formal sector	31.34	11.85	0	100	8,190
Informal sector	31.38	9.05	0	81.80	8,190
Unemployed	13.05	7.73	0	80	8,190
Out of labor force	24.23	7.08	0	58.14	8,190
Earnings					
	Mean	Std. Dev.	Min	Max	Obs
Any Employment	637.89	348.99	60.88	3,582.08	8,190
Formal sector	788.22	439.49	58.67	15,167.10	8,174
Informal sector	491.28	284.28	20	4,941.10	8,172
Non-wage benefits					
Food	38.89	21.06	0	100	8,165
Transport	36.39	25.40	0	100	8,165
Health	20.86	16.41	0	100	8,165

Notes: Each observation is a destination municipality-year cell. Earnings are measured in R\$ of 2012. The informal sector also includes self-employed workers. Non-wage benefits are calculated only for native workers employed in the formal sector.

Table 3: **Comparative characteristics: Migrants vs Natives**

	<i>Migrants</i>		<i>Low-ed. natives</i>		<i>High-ed. natives</i>	
	Mean	S.D.	Mean	S.D.	Mean	S.D.
Age	29.19	10.25	38.43	13.30	33.04	11.14
Number of children	2.13	2.98	3.31	3.07	1.39	1.58
Schooling	4.65	3.96	3.25	2.14	10.90	2.52
Earnings	765.89	1,370.52	783.83	1,516.89	1,994.34	3,300.81
Work less than 40 hours/week	0.11	0.31	0.14	0.34	0.21	0.41
Share of employment	0.67	0.47	0.56	0.50	0.71	0.46
Share of formal employment	0.64	0.48	0.60	0.49	0.81	0.39

Notes: This table compares the characteristics of migrants from the Semiarid region and native individuals in destination municipalities. We use data from the 1991 Census on individuals aged between 18-65 in municipalities covered by the *PNAD* survey. Low-education individuals are those with incomplete elementary schooling. Earnings are measured in R\$ of 2010.

Table 4: **Migration outflows induced by weather shocks**

	(1)	(2)	(3)	(4)	(5)	(6)
Rainfall _{t-1}	-0.099*** (0.028)	-0.092*** (0.028)	-0.093*** (0.028)	-0.092*** (0.028)	-0.093*** (0.028)	-0.096*** (0.030)
Rainfall _{t-2}			0.008 (0.030)	0.022 (0.031)		
Rainfall _{t-3}				0.059** (0.028)		
Rainfall _t					-0.047 (0.031)	
Rainfall _{t+1}						-0.059 (0.036)
Observations	14,400	14,400	14,400	14,400	14,400	14,400
Municipalities	960	960	960	960	960	960
R-Squared	0.461	0.465	0.465	0.466	0.465	0.466
Time dummies	✓	✓	✓	✓	✓	✓
Municipality dummies	✓	✓	✓	✓	✓	✓
Temperature shocks	✓	✓	✓	✓	✓	✓
Baseline × time		✓	✓	✓	✓	✓

Notes: Each observation is an origin municipality-year cell. The dependent variable is the number of individuals who left the origin municipality divided by the total population in the 1991 Census. Rainfall is measured as the log-deviation from the historical average (for the 6 months in the crop growing season). All specifications include controls for temperature shocks, municipality and year fixed effects. Columns (2)-(6) also control for municipality-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. Standard errors are clustered at the grid level. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 5: **Effects of migration on earnings**

	(1)	(2)	(3)
A. Δ log earnings			
Migrant inflow	-1.323*** (0.143)	-1.252*** (0.142)	-0.869*** (0.197)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
B. Δ log earnings, formal sector			
Migrant inflow	-1.005*** (0.171)	-0.929*** (0.169)	-0.593*** (0.198)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
C. Δ log earnings, informal sector			
Migrant inflow	-0.986*** (0.072)	-0.908*** (0.072)	-0.746*** (0.123)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
Time dummies		✓	✓
Baseline $\times$ time			✓

Notes: This table shows origin-level SSIV coefficients on changes in log earnings, by sector. Each observation is an origin municipality-year cell. The informal sector also includes self-employed workers. Column (2) includes time dummies while Column (3) also controls for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. All regressions are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parenthesis. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 6: Effects of migration on employment

	(1)	(2)	(3)
A. Δ employment rate			
Migrant inflow	-0.011 (0.022)	-0.019 (0.022)	-0.018 (0.034)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
B. Δ formal employment rate			
Migrant inflow	-0.312*** (0.025)	-0.317*** (0.025)	-0.126*** (0.037)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
C. Δ informal employment rate			
Migrant inflow	0.301*** (0.029)	0.298*** (0.029)	0.108*** (0.034)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
Time dummies		✓	✓
Baseline $\times$ time			✓

Notes: This table shows origin-level SSIV coefficients on changes in employment rate, by sector. Each observation is an origin municipality-year cell. The informal sector also includes self-employed workers. Column (2) includes time dummies while Column (3) also controls for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. All regressions are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parenthesis. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 7: **Effects of migration on unemployment and participation**

	(1)	(2)	(3)
A. Δ unemployment rate			
Migrant inflow	0.167*** (0.013)	0.176*** (0.013)	0.094*** (0.020)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
B. Δ inactivity rate			
Migrant inflow	-0.155*** (0.022)	-0.157*** (0.022)	-0.077*** (0.029)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
Time dummies		✓	✓
Baseline $\times$ time			✓

Notes: This table shows origin-level SSIV coefficients on changes in unemployment and inactivity rates. Each observation is an origin municipality-year cell. Column (2) includes time dummies while Column (3) also controls for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. All regressions are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parenthesis. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 8: Effects of migration on labor market outcomes, by status in the household

	(1)	(2)	(3)	(4)	(5)
	Employment	Formal	Informal	Unemployment	Inactivity
A. Head					
Predicted inflow	-0.028*	-0.113***	0.085***	0.018*	0.032**
	(0.015)	(0.021)	(0.019)	(0.011)	(0.013)
Observations	11,460	11,460	11,460	11,460	11,460
Municipalities	955	955	955	955	955
B. Non-head					
Predicted inflow	0.010	-0.013	0.024	0.076***	-0.108***
	(0.024)	(0.019)	(0.018)	(0.014)	(0.019)
Observations	11,460	11,460	11,460	11,460	11,460
Municipalities	955	955	955	955	955
Time dummies	✓	✓	✓	✓	✓
Baseline × time	✓	✓	✓	✓	✓

Notes: This table shows origin-level SSIV coefficients on changes in employment (by sector), unemployment, and inactivity rates. Each observation is an origin municipality-year cell. The informal sector also includes self-employed workers. In Panel A we use only individuals identified as the head of the household while in Panel B only those identified as non-head are used. All regressions include time dummies and destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. All regressions are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parenthesis. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 9: Effects of migration on non-wage benefits

	(1)	(2)	(3)
A. Food			
Migrant inflow	-0.336*** (0.062)	-0.369*** (0.063)	-0.687*** (0.086)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
B. Transport			
Migrant inflow	-0.523*** (0.040)	-0.570*** (0.040)	-0.372*** (0.062)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
C. Health			
Migrant inflow	-0.442*** (0.043)	-0.472*** (0.044)	-0.315*** (0.064)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
Time dummies		✓	✓
Baseline × time			✓

Notes: This table shows origin-level SSIV coefficients on changes in the proportions of formal sector workers who receive health insurance, food, or transport subsidies. Each observation is an origin municipality-year cell. Column (2) includes time dummies while Column (3) also controls for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. All regressions are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parenthesis. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 10: Effects of predicted in-migration on employer-provided health insurance

	(1)	(2)	(3)	(4)	(5)
Predicted inflow	-0.015** (0.007)	0.003 (0.002)	-0.004 (0.003)	-0.010** (0.005)	-0.048** (0.022)
Mean of dep. var.	0.0158	0.0131	0.0448	0.0609	0.0758
Observations	4,462,346	4,167,842	138,572	142,100	13,832
Municipalities	682	679	482	608	280
Firms	318,739	297,703	9,898	10,150	988
Time dummies	✓	✓	✓	✓	✓
Firm dummies	✓	✓	✓	✓	✓
Firm size	All firms	1 to 50	51 to 100	101 to 1,000	More than 1,000

Notes: This table shows the reduced form coefficients of changes in the probability of a firm offering health insurance to its employees on the predicted inflow of migrants from the Semiarid region. Each observation is a firm-year cell. The dependent variable is the difference in the dummy variable that is equal to one for every year greater than or equal to the year when the health insurance contract was signed. The regressor is the predicted number of migrants from the Semiarid region in each destination municipality (excluding those in the Semiarid region), measured as a fraction of the native working-age population in 1991. Our sample comprises a balanced panel of all firms included in RAIS during the period. All the regressions are weighted by the number of employees in the firm in 1996. Standard errors clustered at the municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 11: Effects of migration on employment and earnings, by level of MW bite

	A. Δ log earnings					
	(1)	(2)	(3)	(4)	(5)	(6)
	High MW Bite			Low MW Bite		
	Overall	Formal	Informal	Overall	Formal	Informal
Migrant inflow	-1.295*** (0.445)	-0.696 (0.497)	-1.605*** (0.234)	-0.655*** (0.176)	-0.718*** (0.160)	-0.181 (0.135)
Observations	9,840	9,833	9,840	11,460	11,460	11,460
Municipalities	820	820	820	955	955	955
	B. Δ employment rate					
	High MW Bite			Low MW Bite		
	Overall	Formal	Informal	Overall	Formal	Informal
	Overall	Formal	Informal	Overall	Formal	Informal
Migrant inflow	0.002 (0.064)	-0.303*** (0.098)	0.304*** (0.061)	-0.026 (0.035)	-0.079*** (0.024)	0.054 (0.046)
Observations	9,840	9,840	9,840	11,460	11,460	11,460
Municipalities	820	820	820	955	955	955
Time dummies	✓	✓	✓	✓	✓	✓
Baseline $\times$ time	✓	✓	✓	✓	✓	✓

Notes: This table shows origin-level SSIV coefficients on changes in employment rate and log earnings, for each sector and by level of minimum wage bite. Each observation is an origin municipality-year cell. The informal sector also includes self-employed workers. All regressions include time dummies, control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies, and are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parenthesis. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 12: Effects of migration on non-wage benefits, by level of MW bite

	High MW Bite			Low MW Bite		
	Food	Transport	Health	Food	Transport	Health
Migrant inflow	0.070 (0.153)	0.110 (0.126)	0.208** (0.100)	-0.849*** (0.086)	-0.477*** (0.075)	-0.524*** (0.064)
Observations	9,840	9,833	9,840	11,460	11,460	11,460
Municipalities	820	820	820	955	955	955
Time dummies	✓	✓	✓	✓	✓	✓
Baseline × time	✓	✓	✓	✓	✓	✓

Notes: This table shows origin-level SSIV coefficients on changes in the proportions of formal sector workers who receive health insurance, food, or transport subsidies, by level of minimum wage bite. Each observation is an origin municipality-year cell. All regressions include time dummies, control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies, and are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parenthesis. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 13: **Correlation between predicted migration from the Semiarid and other regions**

	(1)	(2)
	Migrant inflow from other regions	
Predicted inflow	0.080 (2.847)	0.081 (2.848)
Observations	8,190	8,190
Municipalities	684	684
Municipality dummies	✓	✓
Time dummies	✓	✓
Baseline $\times$ time		✓

Notes: This table shows destination-level regression coefficients of the observed inflow of migrants from other regions on the predicted number of migrants from the Semiarid, both measured as a fraction of the working-age native population in 1991. Each observation is a destination municipality-year cell. All regressions include municipality and time dummies. Column (2) controls for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. All regressions are weighted by the working-age native population in 1991. Standard errors clustered at the destination municipality level in parenthesis. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 14: Effects of migration on earnings, by industry

	(1)	(2)	(3)
	Agriculture	Manufacturing	Services
Predicted inflow	-0.466 (0.375)	-0.255 (0.188)	-0.831*** (0.187)
Observations	11,447	11,460	11,460
Municipalities	955	955	955
1991 Demographics $\times$ Time	✓	✓	✓
Time dummies	✓	✓	✓

Notes: This table shows origin-level SSIV coefficients on changes in log earnings, by industry. Each observation is an origin municipality-year cell. All regressions include time dummies and control for destination-level 1991 characteristics (log of the working-age native population; shares of population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. All regressions are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parenthesis. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 15: **Long run effects**

	(1)	(2)	(3)
	Overall	Formal	Informal
A. Δ log earnings			
Migrant inflow	-1.111*** (0.312)	-0.658*** (0.253)	-1.570*** (0.265)
Observations	1910	1910	1910
Municipalities	955	955	955
B. Δ employment rate			
Migrant inflow	-0.305*** (0.051)	-0.257*** (0.048)	-0.048 (0.053)
Observations	1,910	1,910	1,910
Municipalities	955	955	955
Time dummies	✓	✓	✓
Baseline $\times$ time	✓	✓	✓

Notes: This table shows origin-level SSIV coefficients of stacked long differences in log earnings and in the employment rate. Each observation is an origin municipality-year cell. The long difference is calculated from 1996-2001 and from 2001-2009. The instrument is the predicted migration accumulated in the same periods, measured as a fraction of the 1991 working-age native population. All regressions include time dummies and destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. All regressions are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parenthesis. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 16: **Long run impacts: non-wage benefits**

	(1)	(2)	(3)
	Food	Transport	Health
Migrant inflow	-0.112 (0.124)	-0.753*** (0.088)	-0.384*** (0.068)
Observations	1,910	1,910	1,910
Municipalities	955	955	955
Time dummies	✓	✓	✓
Baseline $\times$ time	✓	✓	✓

Notes: This table shows origin-level SSIV coefficients of stacked long differences in the proportion of native formal workers receiving non-wage benefits. The long difference is calculated from 1996-2001 and from 2001-2009. The instrument is the predicted migration accumulated in the same periods, measured as a fraction of the 1991 working-age native population. All regressions include time dummies and destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. All regressions are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parenthesis. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table 17: **Effects on migration outflows of natives**

	(1)	(2)	(3)	(4)
Panel A. Low education				
Predicted inflow	1.352 (0.822)	1.273 (0.775)		
Lagged pred. inflow			-0.586 (0.615)	-0.342 (0.450)
Observations	8,190	8,190	8,190	8,190
Municipalities	684	684	684	684
Panel B. High education				
Predicted inflow	0.151 (0.746)	0.109 (0.750)		
Lagged pred. inflow			0.871 (1.186)	0.466 (0.838)
Observations	8,190	8,190	8,190	8,190
Municipalities	684	684	684	684
Municipality dummies	✓	✓	✓	✓
Time dummies	✓	✓	✓	✓
Baseline × time		✓		✓

Notes: This table shows regression coefficients of the number of people leaving the destination areas against the predicted number of migrants from the Semiarid region at the origin municipality level, both measured as a fraction of the native working-age population in 1991. Each observation is a destination municipality-year cell. In Columns (1)-(2) the regressor is the contemporaneous predicted migrant flow, while in Columns (3)-(4) is the same variable lagged one year. All specifications include municipality and time dummies and are weighted by the 1991 native population. Columns (2) and (4) also control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. Standard errors clustered at the destination municipality level in parenthesis. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Internal Migration and Labor Market Adjustments in the Presence of Non-wage Compensation

Raphael Corbi, Tiago Ferraz, and Renata Narita

ONLINE APPENDIX

Appendix A A Simple Model with Informality

In interpreting our findings, we develop a simple extension of a model with informality in which the formal sector has minimum wage but offers non-wage benefits that are frequently observed in the data (e.g. employer-provided health insurance). We follow an extension of the labor market model developed by [Harris and Todaro \(1970\)](#) provided in [Almeida and Carneiro \(2012\)](#). In their model minimum wage and labor legislation are the main institutions behind the existence of a formal and an informal sector. We add non-wage benefits in the formal sector as a source of adjustment of total compensation in the presence of binding minimum wages. We abstract from other sources of frictions, which is explored in much recent work on models of the labor market with monopsony to study immigration effects (e.g. [Amior and Manning \(2020\)](#) and [Amior and Stuhler \(2022\)](#)). They are not needed to understand the mechanism we emphasize, so we proceed with the following model.

Suppose an aggregate output function that combines both formal and informal labor inputs:

$$Y = f(L_f^d, L_i^d, \bar{K}) \quad (\text{A1})$$

where L_f^d and L_i^d are total formal and informal labor, respectively, required to production and $\bar{K}$ the fixed capital stock. $f_{L_f^d L_i^d} \neq 0$ captures production linkages. The wage or the value of a job in the formal and informal sectors are determined by marginal products in each sector, i.e.

$$W_i = \frac{\partial f}{\partial L_i^d} \quad (\text{A2})$$

$$W_f + (1 - t)B = \frac{\partial f}{\partial L_f^d} \geq \underline{W}_f + (1 - t)B \quad (\text{A3})$$

which yield labor demand equations assuming that the minimum wage is binding while non-wage benefits can be optimally chosen in the formal sector.

The labor market with two sectors at the destination can be represented by the following equations:

Formal labor demand	$L_f^d = a - b(\underline{W}_f + (1 - t)B) + cW_i$
Informal labor demand	$L_i^d = d - eW_i + f(\underline{W}_f + (1 - t)B)$
Formal labor supply	$L_f^s = g + h(\underline{W}_f + vB)(1 - U) - iW_i$
Informal labor supply	$L_i^s = j + kW_i - l(\underline{W}_f + vB)(1 - U)$
Equilibrium	$L_f^d = L_f^s(1 - U) = L_f^*$; $L_i^d = L_i^s = L_i^*$
Labor constraint	$L_f^s + L_i^s + O = M$

where W_f and W_i denote wages in the formal and informal sectors, respectively. B are non-wage benefits offered in the formal sector only. For simplicity, we also do not

consider labor taxes or enforcement costs since this is not central in this paper. With the exception of the intercepts of the equations, we assume that all parameters are positive also implying that the two types of labor (formal and informal) are substitutes ($f_{L_f^d L_i^d} < 0$). Employers hiring formal workers can offer benefits (e.g. health insurance and food subsidies) at a cost that is below the wage cost ($t \leq 1$). We assume that workers value such benefits at the rate v , which can be smaller, equal or even larger than 1. The total number of individuals in the economy is M (natives plus migrants), who can either work or search for a job in the formal sector (L_f^s), work in the informal sector (L_i^s), or be out of the labor force (O). Labor markets are competitive, and equilibrium wages and quantities of labor in each sector are determined by the intersection of supply and demand.

We solve for L_f^* , L_i^* , W_i , B and U . The solution to this system is complex so we provide a numerical solution, given the above parametrization. The details of the construction of our numerical example are described in the footnote below.³⁷

In our model, we consider that migration exogenously shifts the supply of workers to the informal and formal sectors. Suppose an equal increase of 10p.p. in the parameters g and j (intercept shifters), so that the total labor force M increases by 20p.p.

Table A1 shows the effects of migration on equilibrium allocations for each economy under two different scenarios: (i) with flexible non-wage benefits, and (ii) with non-wage benefits fixed at the baseline level. Panel A considers our benchmark economy with $c = f = 0.5$ while Panels B and C allow for higher and lower linkages in production across sectors, which is done by imposing $c = f = 0.7$ and $c = f = 0.3$, respectively. All baseline economies in column (1) are normalized to 100.

In columns (2) and (3) we show the effects of introducing migration in each scenario. Panel B shows that in the benchmark economy with medium production linkages across sectors and varying benefits (column (2)), migration increases unemployment and informal employment, and drops non-wage benefits and informal sector wages. Formal employment is unchanged. With benefits fixed at the baseline level (column (3)), the overall impacts on employment and informal wages are relatively higher since benefits do not adjust. Moreover, formal employment declines.

Panel C considers an economy with higher linkages, the new equilibrium reflects a better adjustment of informal wages and non-wage benefits to the fact that there are wage rigidities in the formal sector, despite a lower value of non-wage benefits. Consequently, the effects of migration with flexible non-wage benefits are lower than in the economy in Panel B. With fixed benefits, the impacts are even lower or nonexistent for most outcomes given the fast degree of market integration.

³⁷We set our benchmark at $a = d = 1$, $g = j = 0$, $b = e = h = k = 1$ and $c = f = i = l = 0.5$. The slope restrictions are consistent with integrated formal and informal sectors but we do consider that own effects are likely larger than cross-effects determining demand and supply of labor in each sector. We also consider that offering benefits is 50% cheaper to firms consistent with fiscal exemptions on such benefits ($t = 0.5$) and that workers value non-wage benefits less than wages with $v = 0.5$, motivated by lack of liquidity or pension accumulation. Finally, given the above parametrization, we set the minimum wage $\underline{W}_f$ at 1, $O = 0$, and $M = 1$.

Table A1: **Effects of introducing migration in a labor market with informality**

	(1)	(2)	(3)
	benchmark	migration with B^*	migration with $\bar{B}$
Panel A: Low linkages			
L_f^*	100	100	88
L_i^*	100	115	124
W_i	100	93	89
B	100	33	100
U	100	128	156
Panel B: Medium linkages			
L_f^*	100	100	89
L_i^*	100	109	119
W_i	100	94	90
B	100	45	100
U	100	183	283
Panel C: High linkages			
L_f^*	100	100	100
L_i^*	100	103	100
W_i	100	96	100
B	100	96	100
U	100	98	97

Under lower linkages, Panel A shows that the qualitative results of Panel B are kept. However, migration induces a larger fall in informal sector wages and formal sector benefits with flexible formal sector benefits. With fixed benefits, the results in column (3) show that migration has now a negative impact on formal employment, as expected since formal firms cannot adjust benefits after the supply shock from migration. Consequently, unemployment also increases more under fixed benefits in such economy.

Appendix B Migrant flows from the Semiarid region

In this section, we discuss in more detail our measure of migration between cities and how we structure a yearly panel dataset from the 2000 and 2010 censuses.

B.1 Migration from the Semiarid region

In every round of the Census, there are two questions that allow us to track the migrants and establish their municipalities of origin and destination, as well as the year when they moved.

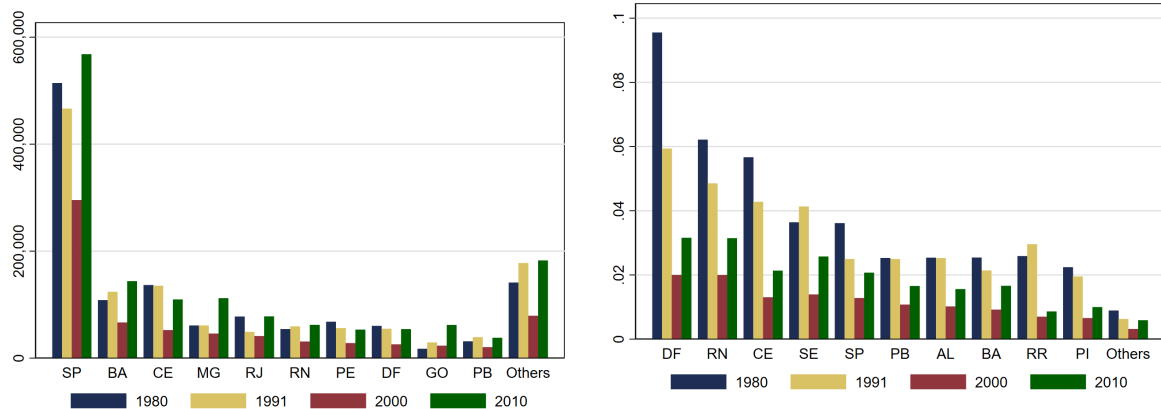
First, in the 2010 Census, respondents were asked for how many years they had lived in the current municipality (from one up to ten). With this variable, we are able to calculate the year when the individual has migrated. We consider a migrant an individual who moved to the current municipality in the previous ten years. In the 2000 Census, interviewees were asked about the municipality where they were living five years ago, instead of the last place where they lived so we can only identify migrants who came as far as 1996. This is not a major concern in our analysis as 1996 is the first year for which *PNAD* data - the source from which we draw labor market outcomes information - is available.

Second, they were asked what was the municipality where they lived before. Thus, if an individual has migrated from an origin municipality in the Semiarid region, she will be counted as a Semiarid migrant. A limitation is that we can only track one origin location for each person, probably the last municipality where she lived.

The Semiarid region has always been an important source of migrants for the rest of the country. Figure B1 shows that these migrants tend to be historically concentrated in some states. São Paulo alone harbored over 30 percent of the people arriving from the Semiarid in the last four decades. However, in relative terms, incoming migrants represented a population increase of above 2% for the top 10 receiving states.

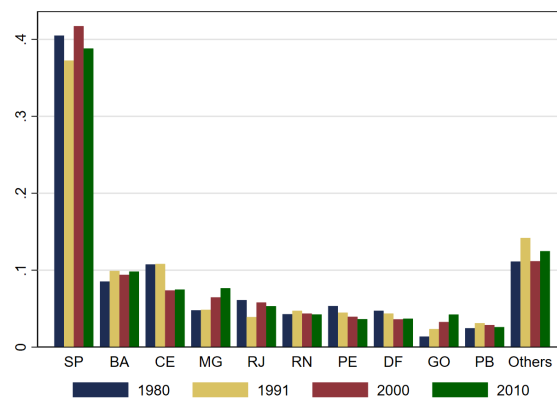
Table 3 compares migrants to low and high-education natives. Migrants are slightly more educated and earn slightly less than less-educated natives. They also have a similar likelihood of working part-time and being in the formal sector when compared to low-education natives. On the other hand, high-education natives are more likely to work in the formal sector and have considerably higher pay. Table B1 shows that top occupations for migrants (e.g. typically bricklayer for men, domestic worker for women) are also top occupations for low-education natives, but not for the skilled. Also, the same five industries that concentrate over 80% of working migrants also employ a similar share of low-education workers (see Table B2).

Figure B1: Top destinations for migrants from the Semi-arid region



(a) Absolute number of Semi-arid's migrants

(b) Semi-arid's migrants as a fraction of total population



(c) Semi-arid's migrants as a share of total migration

Notes: This figure presents the main destination states chosen by migrants from the Semi-arid region. Panel (a) shows the absolute number of migrants leaving the Semi-arid region to non-Semi-arid areas. Panel (b) presents the same inflow measured as a fraction of the total population in the state while in Panel (c) that number is measured as a share of the total number of migrants in each state. In each panel, states are ranked by the respective average across years. *Data source:* Census microdata (IBGE).

Table B1: **Main occupations for employed people: Migrants vs Natives**

	Position	Occupation	Share of em- ployment	Cumulative
<i>Migrants</i>	1	Domestic worker	13.8	13.8
	2	Bricklayer	9.6	23.4
	3	Non-specified occupations	9.1	32.5
	4	Salesperson	9.1	41.5
	5	Rural worker	3.6	45.2
	6	Janitor	3.0	48.2
	7	Office assistant	2.6	50.8
	8	Tailor	2.5	53.3
	9	Driver	2.3	55.5
	10	Security guard	2.0	57.5
<i>Low-ed. natives</i>	1	Rural worker	10.8	10.8
	2	Bricklayer	8.2	19.0
	3	Salesperson	8.1	27.0
	4	Domestic worker	7.8	34.8
	5	Non-specified occupations	6.0	40.8
	6	Driver	5.7	46.5
	7	Janitor	3.6	50.1
	8	Tailor	2.9	53.0
	9	Cook	1.7	54.7
	10	Mechanic	1.7	56.5
<i>High-ed. natives</i>	1	Salesperson	8.9	8.9
	2	Office assistant	7.9	16.7
	3	Non-specified occupations	4.4	21.1
	4	Tradesperson	3.1	24.2
	5	Secretary	3.1	27.2
	6	Driver	2.6	29.9
	7	Office supervisor	2.6	32.5
	8	Military	2.0	34.5
	9	Teacher	2.0	36.4
	10	Nurse	1.8	38.2

Notes: This table presents the top ten occupations for workers in the destination municipalities, using data from the 1991 Census.

Table B2: **Main industries for employed people: Migrants vs Natives**

	Position	Industry	Share of em- ployed	Cumulative
<i>Migrants</i>	1	Hospitality	31.0	31.0
	2	Manufacturing	19.8	50.8
	3	Retail	14.3	65.1
	4	Construction	13.0	78.2
	5	Agriculture/Mining	5.6	83.7
	6	Health/Education	5.4	89.1
	7	Transport/Communication	4.0	93.1
	8	Other Services	2.5	95.5
	9	Public Sector	2.5	98.0
	10	Professional Services	2.0	100.0
<i>Low-ed. natives</i>	1	Hospitality	25.5	25.5
	2	Manufacturing	18.8	44.3
	3	Agriculture/Mining	14.8	59.2
	4	Retail	12.6	71.8
	5	Construction	10.9	82.7
	6	Transport/Communication	6.0	88.7
	7	Health/Education	4.9	93.6
	8	Public Sector	3.1	96.7
	9	Professional Services	1.9	98.5
	10	Other Services	1.5	100.0
<i>High-ed. natives</i>	1	Health/Education	18.8	18.8
	2	Manufacturing	17.5	36.3
	3	Retail	16.8	53.1
	4	Hospitality	12.0	65.1
	5	Public Sector	9.2	74.3
	6	Professional Services	7.4	81.7
	7	Other Services	6.8	88.5
	8	Transport/Communication	4.9	93.3
	9	Agriculture/Mining	3.5	96.9
	10	Construction	3.1	100.0

Notes: This table presents the top ten industries for workers in the destination municipalities, using data from the 1991 Census.

Appendix C Weather shocks and predicted migration

In this section, we discuss the weather data and provide further details about how we construct our instrument. We also show that our results are robust to an alternative measure of weather shocks.

C.1 Weather data

Our main source for weather data comes from the CRUTS v4, a gridded dataset produced by the Climatic Research Unit at the University of East Anglia (Harris et al., 2020). It provides information on monthly precipitation and temperature covering the whole globe (except Antarctica) from 1901 to 2018. The grid resolution is $0.5^\circ \times 0.5^\circ$ (around 56km^2) and is created by interpolation from ground-based weather stations around the world.

We use the R package ‘geobr’ (Carabetta et al., 2020) to download the shapefile of Brazilian municipalities and georeference the coordinates from each municipality’s centroid. Then, for each municipality, we find the grid’s four points that are closest to its centroid and calculate the average level of precipitation and temperature from these points, weighted by the inverse distance to the centroid.

This procedure results in a dataset of monthly averages of precipitation and temperature for each municipality, from 1901 to 2010, which we aggregate in yearly measures. Precipitation is defined as the sum of monthly levels and temperature as the average. For each municipality in the Semiarid region, we calculate the historical mean from both variables and take the natural logarithm of these variables (both levels and long-term averages).

Finally, our weather shock variables are defined as

$$Rainfall_{ot} = \ln \left(\sum_{\tau \in \{GS\}} r_{ort} \right) - \ln(\bar{r}_o) \quad (C1)$$

where r_{ort} is the rainfall in the municipality of origin o in month τ of year t , and $\bar{r}_o$ is the municipality’s historical average precipitation for the same months. The index τ covers the 6-month growing season (GS). Temperature is calculated in a similar way, but using the average instead of summation to create yearly data. In our main specifications, we use data from the Semiarid’s growing season (from November to April), but results are very similar when we use the full year (see Table C1).

C.2 Alternative measures of weather

One possible concern about our measure of weather is that we focus on rainfall levels, controlling for temperature variation, to predict the flow of migrants leaving the Semiarid region. This may be problematic because we cannot account for the presence of groundwater or any other factors that influence water balance. To circumvent this issue

Table C1: **Migration outflows induced by weather shocks (12 months)**

	(1)	(2)	(3)	(4)	(5)	(6)
Rainfall _{t-1}	-0.126*** (0.033)	-0.107*** (0.033)	-0.111*** (0.034)	-0.117*** (0.034)	-0.112*** (0.033)	-0.109*** (0.033)
Rainfall _t			-0.015 (0.039)	-0.029 (0.041)	-0.014 (0.039)	
Rainfall _{t-2}			0.037 (0.038)	0.059 (0.039)		
Rainfall _{t-3}				0.047 (0.033)		
Rainfall _{t+1}						-0.068* (0.037)
Observations	14,400	14,400	14,400	14,400	14,400	14,400
Municipalities	960	960	960	960	960	960
R-Squared	0.461	0.465	0.465	0.466	0.465	0.466
Time dummies	✓	✓	✓	✓	✓	✓
Municipality dummies	✓	✓	✓	✓	✓	✓
Temperature shocks	✓	✓	✓	✓	✓	✓
Baseline × time		✓	✓	✓	✓	✓

Notes: Each observation is a municipality-year cell. The dependent variable is the number of individuals who left the origin municipality divided by the total population in the 1991 Census. Rainfall is measured as log-deviation from the historical average. All specifications include controls for temperature shocks, municipality and year fixed effects. Columns (2)-(6) control for municipality-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. Standard errors are clustered at the grid level. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

we use the Standardized Precipitation Evapotranspiration Index (SPEI) developed by [Vicente-Serrano et al. \(2010\)](#). The SPEI is a measure that accounts for both precipitation and potential evapotranspiration, providing a measure of water balance in a given period. This index also captures deviations from the historical average (1905-2018) in the net water need for a given location. An SPEI value of -1 means that the precipitation level is one standard deviation below the historical level needed to maintain the balance given the potential evapotranspiration. According to [Vicente-Serrano et al. \(2010\)](#), the SPEI is particularly useful to detect, monitor, and explore the consequences of global warming on drought conditions. We repeat the first step in our procedure to construct an instrument for in-migration, using the SPEI instead of rainfall and temperature shocks. Once again, we calculate the average SPEI for the 6-month growing season. Table [C2](#) shows that we can also use this measure to predict the out-migration rate from the origin municipalities in the Semiarid region, although the estimates are noisier than those in Table [4](#).

We estimate the same specification of column 3 in Tables [5](#), [6](#) and [9](#) using this new instrument and show in Tables [C3](#) and [C4](#) that the results are very similar.

Table C2: **Migration outflows induced by weather shocks: Standardized Precipitation Evapotranspiration Index (SPEI)**

	(1)	(2)	(3)	(4)	(5)	(6)
$SPEI_{t-1}$	-0.041*** (0.016)	-0.036** (0.016)	-0.037** (0.016)	-0.035** (0.016)	-0.036** (0.016)	-0.037** (0.016)
$SPEI_{t-2}$			0.040** (0.017)	0.040** (0.017)		
$SPEI_{t-3}$				0.014 (0.018)		
$SPEI_t$					-0.012 (0.015)	
$SPEI_{t+1}$						-0.011 (0.016)
Observations	14,400	14,400	14,400	14,400	14,400	14,400
Municipalities	960	960	960	960	960	960
R-Squared	0.461	0.465	0.465	0.465	0.465	0.465
Time dummies	✓	✓	✓	✓	✓	✓
Municipality dummies	✓	✓	✓	✓	✓	✓
Demographics		✓	✓	✓	✓	✓

Notes: Each observation is a municipality-year cell. The dependent variable is the number of individuals who left the origin municipality divided by the total population in the 1991 Census. All specifications include municipality and year fixed effects. Column (6) also controls for municipality-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. Drought severity measures are the quartiles of the Aridity Index z-score. Standard errors are clustered at the grid level. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table C3: Effects of migration on earnings and employment, using SPEI to predict out-migration

	(1)	(2)	(3)
	Overall	Formal	Informal
A. Change in log earnings			
Migrant inflow	-0.844*** (0.197)	-0.562*** (0.198)	-0.730*** (0.122)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
B. Change in employment rate			
Migrant inflow	-0.010 (0.034)	-0.118*** (0.037)	0.108*** (0.033)
Observations	11,460	11,460	11,460
Municipalities	955	955	955
Time dummies	✓	✓	✓
Baseline $\times$ time	✓	✓	✓

Notes: This table shows origin-level SSIV coefficients of regressions on log earnings and on the employment rate for native workers. Each observation is an origin municipality-year cell. The instrument for the migrant inflow is calculated using the SPEI to predict out-migration distributed by the 1991 share of Semiarid migrants. All regressions include time dummies and destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. All regressions are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parenthesis. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table C4: Effects of migration on non-wage benefits, using SPEI to predict out-migration

	(1)	(2)	(3)
	Food	Transport	Health
Migrant inflow	-0.717*** (0.087)	-0.398*** (0.063)	-0.332*** (0.063)
Observations	11460	11460	11460
Municipalities	955	955	955
Time dummies	✓	✓	✓
Baseline $\times$ time	✓	✓	✓

Notes: This table shows origin-level SSIV coefficients of regressions on the proportion of native formal sector workers receiving non-wage benefits. Each observation is an origin municipality-year cell. The instrument for the migrant inflow is calculated using the SPEI to predict out-migration distributed by the 1991 share of Semiarid migrants. All regressions include time dummies and destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. All regressions are weighted by the working-age native population in 1991. Standard errors clustered at the origin municipality level in parenthesis. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Appendix D Shift-share instrument (SSIV)

In this section, we derive the origin-level SSIV estimator, and present and discuss the identifying assumptions needed to produce a consistent estimator of the effects of the inflow of migrants from the Semiarid region on labor markets in the destination municipalities.

We start from the structural equation 1. To simplify notation we omit the time subscript t . By the Frisch-Waugh-Lovell Theorem, we can re-write it as

$$y_d^\perp = \beta m_d^\perp + \varepsilon_d^\perp \quad (\text{D1})$$

where all $y_d^\perp$ is the vector of outcomes, $m_d^\perp$ ³⁸ is the observed number of Semiarid's migrants who entered the destination municipality d and $\varepsilon_d^\perp$ is a structural residual. All variables are residualized to remove the effects from the covariates.

In equation 4 we defined the shift-share instrumental variable (SSIV) as

$$\hat{m}_d = \sum_{o=1}^O s_{od} \frac{\widehat{M}_o}{P_d} \quad (\text{D2})$$

where s_{od} is the share of migrants from origin municipality o who lived in the destination area d in 1991 and $\widehat{M}_o$ is the predicted number of migrants leaving the Semiarid region driven by weather shocks.³⁹

The more traditional approach would be to estimate β using $\hat{m}_d$ as an instrument for the endogenous migrant inflow $m_d^\perp$. In such case, we would have

$$\hat{\beta} = \frac{\sum_d \hat{m}_d y_d^\perp}{\sum_d \hat{m}_d m_d^\perp} \quad (\text{D3})$$

By the definition of $\hat{m}_d$ in equation D2 and switching the order of the summation,

$$\hat{\beta} = \frac{\sum_d \left(\sum_o s_{od} \frac{\widehat{M}_o}{P_d} \right) y_d^\perp}{\sum_d \left(\sum_o s_{od} \frac{\widehat{M}_o}{P_d} \right) m_d^\perp} = \frac{\sum_o \widehat{M}_o \left(\sum_d s_{od} \frac{y_d^\perp}{P_d} \right)}{\sum_o \widehat{M}_o \left(\sum_d s_{od} \frac{m_d^\perp}{P_d} \right)} = \frac{\sum_o s_o \widehat{M}_o \bar{y}_o}{\sum_o s_o \widehat{M}_o \bar{m}_o} \quad (\text{D4})$$

where $\bar{y}_o = \frac{\sum_d s_{od} \frac{y_d^\perp}{P_d}}{\sum_d s_{od}}$ is a weighted average of the residualized outcome, normalized by the population, which uses as weights the destination's average exposure to the shocks $s_o = \sum_d s_{od}$. The same result is valid for the endogenous variable $m_d^\perp$, meaning that we can estimate the following IV regression at the origin municipality level:

$$\bar{y}_o = \beta \bar{m}_o + \bar{\varepsilon}_o \quad (\text{D5})$$

³⁸In order to facilitate the interpretation of the coefficients we normalize this measure dividing by the working-age native population in 1991, which means $m_d = \frac{M_d}{P_d}$

³⁹The same normalization is applied in the predicted inflow.

using the predicted number of migrants from the Semiarid region, $\widehat{M}_o$, as instrumental variable and weighting by the average exposure s_o .

This derivation is almost identical to that presented by [Borusyak et al. \(2021\)](#), except for the fact that we need to divide both variables by the predetermined population. But, this is only a normalization using the destination's native population and the equivalence result shows that the parameter β can be estimated at the level of the identifying variation, which in our case is the origin municipality hit by weather shocks.

Table D1: **SSIV First Stage**

	(1)	(2)	(3)
First stage coefficient	0.912*** (0.015)	0.910*** (0.015)	0.925*** (0.019)
F-statistic	3,462	3,464	2,275
Observations	11,460	11,460	11,460
Municipalities	955	955	955
Effective sample size	7,301	7,301	7,301
Time dummies		✓	✓
Baseline × time			✓

Notes: This table shows the SSIV first stage coefficients of the origin-level weighted average of the endogenous inflow of migrants at the destinations against the predicted number of migrants from the Semiarid region. Each observation is an origin municipality-year cell. The F-statistic is calculated as the square of the coefficient t-statistic (see [Borusyak et al., 2021](#)). The effective sample size is the inverse of the HHI of the origin-level exposure. Column (2) includes time dummies while Column (3) also controls for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. Regressions are weighted by the working-age native population in 1991. Standard errors cluster by the municipality of origin in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Appendix E Spatial correlation in weather shocks

Weather events are likely correlated across space. Figure 12 shows that precipitation levels in the Semiarid are similar among nearby municipalities. Potentially, this could invalidate the consistency of our estimator given by *Assumption 2 (Many uncorrelated shocks)* discussed in Appendix D. Here we investigate this issue by re-constructing our instrument according to different degrees of aggregation of regions of origin of a migrant - such as a microregion or a mesoregion - instead of a municipality. IBGE (1990) defines microregions as “groups of economically integrated municipalities sharing borders and structure of production”. Mesoregions are collections of microregions of which not all municipalities share borders.⁴⁰ Brazil has 5,565 municipalities, 361 micro, and 87 mesoregions overall. The Semiarid has 960 municipalities, 137 micro, and 35 mesoregions.

The intuition behind this exercise is that even if weather shocks are spatially correlated among contiguous municipalities, such a correlation should decrease as we consider larger areas. Table E2 displays Moran’s index of spatial correlation of rainfall shocks for each of the three geographic aggregates in columns 1-3.⁴¹ As expected, neighboring municipalities display a correlation above 0,94, but it decreases rapidly as we aggregate up to micro and meso regions, to 0,16 and 0,07, respectively.

Table E2 also shows the association between rainfall shocks and migration outflows. Column 1 is identical to Table 4 for reference. Columns 2 and 3 report almost identical point estimates and precision, indicating that we do not lose any significant information by aggregating origin areas. Next we estimate our main specification from Column (3) in Tables 5,6 and 9 using instruments corresponding to micro and mesoregion-level aggregation. Tables E3-E5 show that our results associating migration and earnings, employment and non-wage benefits are very similar to the municipality-level estimates, although standard errors increase substantially, as one would expect considering that there are fewer units from which we can leverage variation. All those results indicate that spatial correlation among rainfall shocks in origin municipalities is not a source of relevant bias in our setting.

⁴⁰Table E1 reports summary statistics of our main variables for both levels of aggregation.

⁴¹Moran’s I is calculated according to the following formula:

$$I = \frac{1}{\sum_i \sum_j w_{ij}} \times \frac{\sum_i \sum_j w_{ij} (y_i - \bar{y})(y_j - \bar{y})}{\frac{1}{N} (\sum_i (y_i - \bar{y})^2)} \quad (E1)$$

Essentially, it is a correlation coefficient weighted by an appropriate matrix that models how different units are related across space. We use a row-standardized contiguity matrix with the queen criterion, meaning that two localities i and j sharing either borders or vertices are considered ‘neighbors’ and the entry w_{ij} has a positive value. Row-standardization ensures that weights are positive and no greater than 1. Non-adjacent pairs receive a zero weight. As discussed by Beenstock et al. (2019), Moran’s I can be calculated for each period and averaged out with panel data.

Table E1: **Summary statistics: Micro- and meso-regions in the Semiarid**

Panel A - Micro-regions	Mean	Std. Dev.	Min	Max	Obs
Rainfall shock	-0.01	0.20	-0.70	0.47	2,055
Temperature shock	0.01	0.01	-0.01	0.05	2,055
Out-migration	1,500.70	1,371.95	6.00	9,685.00	2,055
Out-migration rate (p.p.)	1.08	0.41	0.12	3.12	2,055
Population	148,981.55	128,183.19	4,968	752,719	2,055
Area	7,150.16	7,857.60	84.94	55,358.33	2,055
Number of municipalities	8.20	4.56	2.00	26.00	2,055
Panel B: Meso-regions	Mean	Std. Dev.	Min	Max	Obs
Rainfall shock	-0.02	0.20	-0.69	0.44	525
Temperature shock	0.01	0.01	-0.01	0.05	525
Out-migration	5,874.18	5,766.16	51.00	34,800.00	525
Out-migration rate (p.p.)	1.08	0.37	0.24	2.32	525
Population	583,156.36	524,776.40	15,499	2,349,152	525
Area	27,986.83	30,649.61	84.94	124,505.71	525
Number of municipalities	37.20	21.51	10.00	118.00	525

Notes: Rainfall is measured in mm. Temperature is measured in degrees Celsius. Migration outflow (inflow) rate is the share of migrants over the local (native) population. The area is measured in km².

Table E2: **Migration outflows induced by weather shocks according to different aggregation levels**

	(1)	(2)	(3)
	Municipality	Micro-region	Meso-region
Rainfall _{<i>t</i>-1}	-0.099*** (0.028)	-0.094*** (0.032)	-0.099*** (0.025)
Observations	14,400	2,055	525
Origins	960	137	35
R-Squared	0.461	0.764	0.866
Moran's I	0.947	0.158	0.075
Time dummies	✓	✓	✓
Origin dummies	✓	✓	✓
Temperature shocks	✓	✓	✓

Notes: Each observation is a region-year cell. The dependent variable is the number of individuals who left the origin region divided by the total population in the 1991 Census. Rainfall is measured as log-deviation from the historical average. All specifications include controls for temperature shocks, municipality and year fixed effects. Moran's I show the spatial correlation in rainfall shocks among origin regions. Standard errors are clustered at the respective region level. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table E3: Effects of migration on earnings
according to different aggregation levels

	(1)	(2)	(3)
A. Δ log earnings			
	Municipality	Micro-region	Meso-region
Predicted inflow	-0.869*** (0.197)	-0.846*** (0.302)	-0.871 (0.550)
Observations	11,460	1,644	420
Regions	955	137	35
B. Δ log earnings, formal sector			
	Municipality	Micro-region	Meso-region
Predicted inflow	-0.593*** (0.198)	-0.558* (0.290)	-0.556 (0.527)
Observations	11,460	1,644	420
Regions	955	137	35
C. Δ log earnings, informal sector			
	Municipality	Micro-region	Meso-region
Predicted inflow	-0.746*** (0.123)	-0.745*** (0.201)	-0.769** (0.343)
Observations	11,460	1,644	420
Regions	955	137	35
Demographics	✓	✓	✓
Time dummies	✓	✓	✓

Notes: This table shows origin-level SSIV coefficients on changes in log earnings, by sector. Each observation is an origin municipality-year cell. The informal sector also includes self-employed workers. All specifications include time and control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. Column (1) replicates the same results from Column (3) of Table 5. In columns (2) and (3) we aggregate the origin-level shocks at the micro- and meso-region levels, respectively. All regressions are weighted by native working-age population in 1991. Standard errors clustered at the respective aggregation level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table E4: **Effects of migration on employment
according to different aggregation levels**

	(1)	(2)	(3)
A. Δ employment rate			
	Municipality	Micro-region	Meso-region
Predicted inflow	-0.018 (0.034)	-0.003 (0.058)	0.007 (0.091)
Observations	11,460	1,644	420
Regions	955	137	35
B. Δ formal employment rate			
	Municipality	Micro-region	Meso-region
Predicted inflow	-0.126*** (0.037)	-0.117** (0.055)	-0.125 (0.098)
Observations	11,460	1,644	420
Regions	955	137	35
C. Δ informal employment rate			
	Municipality	Micro-region	Meso-region
Predicted inflow	0.108*** (0.034)	0.114** (0.057)	0.133 (0.095)
Observations	11,460	1,644	420
Regions	955	137	35
Demographics	✓	✓	✓
Time dummies	✓	✓	✓

Notes: This table shows origin level SSIV coefficients of Δ the proportions of employed natives, by sector. Each observation is an origin municipality-year cell. The informal sector also includes self-employed workers. All specifications include time and control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. Column (1) replicates the same results from Column (3) of Table 6. In columns (2) and (3) we aggregate the origin-level shocks at the micro- and meso-region levels, respectively. All specifications use the same set of controls defined in Table 6. All regressions are weighted by native working-age population in 1991. Standard errors clustered at the municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table E5: Effects of migration on non-wage benefits
according to different aggregation levels

	(1)	(2)	(3)
A. Food			
	Municipality	Micro-region	Meso-region
Predicted inflow	-0.687*** (0.086)	-0.658*** (0.134)	-0.688*** (0.216)
Observations	11,460	1,644	420
Regions	955	137	35
B. Transport			
	Municipality	Micro-region	Meso-region
Predicted inflow	-0.372*** (0.062)	-0.305*** (0.104)	-0.290* (0.157)
Observations	11,460	1,644	420
Regions	955	137	35
C. Health			
	Municipality	Micro-region	Meso-region
Predicted inflow	-0.315*** (0.064)	-0.289*** (0.097)	-0.312* (0.173)
Observations	11,460	1,644	420
Regions	955	137	35
Demographics	✓	✓	✓
Time dummies	✓	✓	✓

Notes: This table shows origin level SSIV coefficients of change in the proportions of formal sector workers who receive health insurance, food, or transport subsidies. Each observation is an origin municipality-year cell. All specifications include time and control for destination-level 1991 characteristics (log of the working-age native population; shares of the population aged 15-25, 26-50, 51-65 and older than 65; share of the non-white population; share of the population with a college education; share of women in the total and employed populations; shares of employment in agriculture and manufacturing; logs of the average household income and size; and the shares of households with access to electricity and piped water) interacted with time dummies. Column (1) replicates the same results from Column (3) of Table 9. In columns (2) and (3) we aggregate the origin-level shocks at the micro- and meso-region levels, respectively. All regressions are weighted by native working-age population in 1991. Standard errors clustered at the municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.